DOMAIN 6
OCCUPATIONAL HEALTH AND APPLIED SCIENCE (QUESTIONS & ANSWERS)

1622. 5 workers are exposed to airborne lead at 8-hr TWA's of 27 ag/m^3, 42 ag/m^3, 54 ag/m^3, 24 ag/m^3, and 60 ag/m^3. What is the mean exposure?

1) 41 ag/m^3
2) 42 ag/m^3
3) 46 ag/m^3
4) 40 ag/m^3

Ans.1 In order to get the mean average exposure for the five workers, sum the exposures for all workers first.
(27 + 42 + 54 + 24 + 60) ag/m^3 = 207 ag/m^3
Then calculate the mean exposure by dividing the sum by the total number of workers.
207 ag/m^3 / 5 workers = 41.4 a/m^3 per worker

1623. Six workers are exposed to methylene chloride at the following levels (based on an 8-hr TWA): 27 ppm, 31 ppm, 12 ppm, 86 ppm, 27 ppm, and 40 ppm. What is the statistical mode of their exposures?

1) 37 ppm
2) 31 ppm
3) 27 ppm
4) 29 ppm

Ans.3 The mode is the number in a set of numbers that appears most frequently. 27 ppm is correct because it appears most frequently in this set of numbers.

1624. TLV-C refers to an airborne exposure concentration:

1) That should never be exceeded
2) Permitted for a 30 minute exposure
3) That is below the TLV-TWA
4) That is an emergency exposure level

Ans.1 The TLV-C (threshold limit value-ceiling) should never be exceeded.

1625. Which of the following is the LEAST likely property of activated carbon that makes it a good sample collecting medium?

1) Large surface area
2) Ease of stripping collected samples
3) High adsorber for gases, vapors
4) High polarity

Ans.4 Activated carbon is extremely porous and has a non-polar surface. It adsorbs molecules to its surface readily. When bathed in a non-polar solvent such as carbon disulfide, the adsorbed molecules are easily removed and dissolved into the solvent.

1626. Silicosis is a dust disease from exposure to:

1) Glass
2) Crystalline SiO_2
3) Amorphous silica
4) Aluminum silicate

Ans.2 All of the answers contain the element silicon. However, only crystalline silicon dioxide (SiO_2) causes the disease silicosis. Dust from glass, amorphous silica, and aluminum silicate are considered nuisance dusts and have minimal long term biological effects.

1627. Vibration exposure is monitored by measuring all of the following EXCEPT:

1) Acceleration
2) Decibels
3) Frequency
4) Duration of exposure

Ans.2 Decibels are units used for measuring noise exposure. Vibration is measured in acceleration, frequency, and duration.

1628. The octave band center frequencies that most closely correlate with human speech are:

1) 750 Hz, 1,500 Hz, 3,000 Hz
2) 250 Hz, 500 Hz, 1,000 Hz
3) 1,000 Hz, 1,500 Hz, 2,000 Hz
4) 500 Hz, 1,000 Hz, 2,000 Hz

Ans.4 The center frequencies of the octave bands that most closely correlate with human speech are 500 Hz, 1,000 Hz, and 2,000 Hz.

1629. Numerical values that are based on procedures to determine the amount of hazardous materials the human body has absorbed (by measuring the actual materials or their metabolic products in tissue, fluid, or exhaled air) is called a:

1) BCM
2) BLD
3) BEI
4) BP

Ans.3 BEI's (Biological Exposure Indexes) are numerical values based on procedures to determine the amount of a material the human body absorbs by measuring the actual materials or their metabolic products in tissue, fluid, or exhaled air.

1630. What is the percentage of all occupational illnesses accounted for by cumulative trauma disorders and skin diseases/disorders?

1) 30%
2) 40%
3) 50%
4) 70%

Ans.4 Cumulative trauma disorders and skin diseases and disorders account for over 70% of all occupational illnesses.

1631. 3.7×10^{10} disintegrations per second equals _____ disintegrations per minute.

1) 2.2×10^{18}

2) 6.2×10^{8}
3) 1.0×10^{7}
4) 1.3×10^{14}

Ans.1 To calculate the number of disintegrations per minute, simply multiply the number of disintegrations per second by 60 sec/min.
60 sec/min x 3.7×10^{10} dis/sec = 2.22×10^{12} dis/min

1632. The explosimeter, combustible gas indicator, and vapotester operate on what principle?

1) Adsorption
2) Absorption
3) Beilstein reaction
4) Heat of combustion

Ans.4 All of the listed instruments cause an oxidation to take place and make a measurement based on the heat released.

1633. The principal of the rotameter is similar to:

1) Venturi meter
2) Variable orifice meter
3) Pitot tube
4) Anemometer

Ans.2 As the float rises, the annular width varies in the rotameter. An analogous condition is present when the orifice size is changed in a variable orifice meter.

1634. The term self-accelerating decomposition temperature (SADT), is defined as:

1) The maximum temperature of a liquid at which it gives off vapor sufficient to form an ignitable mixture with air

2) The minimum temperature of a liquid at which it gives off vapor sufficient to form an ignitable mixture with air
3) The ambient temperature at which a self-sustaining chemical reaction of stored organic peroxides will occur
4) The rate of decomposition of certain organic peroxides when stored at very high temperatures

Ans.3 The SADT is determined for stored organic peroxides. It is the ambient temperature at which a run-away reaction will likely occur.

1635. In an automated man-machine system, man acts as a:

1) Power source and controller
2) Controller
3) Power source
4) Monitor

Ans.4 Since the man-machine system is automated, the man acts as a monitor.

1636. The fall season has arrived at a New England office facility and with it, a rash of HVAC complaints that were not recorded during the summer. Which troubleshooting step should be first?

1) Determine if the dampers are operating properly
2) Check the inside/outside temperature controls
3) Determine if the air distribution is balanced to specifications
4) Check the make-up air system, or damper

Ans.4 During the warm months of the year, large volumes of air of relatively high velocities are welcomed by the workers. During the fall or winter months, care must be taken to make sure that air blowing over the occupants is kept within acceptable values (comfort conditions) suitable for the work situation.

1637. The Brief and Scala Model refers to:

1) Biologically-derived airborne contaminants
2) Unusual work schedules
3) TLV's for mixtures
4) Skin notation

Ans.2 The Brief and Scala Model refers to unusual work schedules. The model reduces the TLV proportionately for increased exposure time and reduced non-exposure time. It is generally intended to apply to work schedules longer than 8 hours/day or 40 hours/week.

1638. Immediate corrective measures should be taken when the atmospheric concentration of:

1) Carbon dioxide is less than 2.0%
2) Carbon dioxide is greater than 3.0%
3) Carbon monoxide is less than 0.1%
4) Oxygen is greater than 18% but less than 20%

Ans.2 Immediate corrective measures should be taken when the atmospheric concentration of carbon dioxide is greater than 3.0%.

1639. A survey was conducted to determine the causes of occupational dermatitis. Exposure to which of the following materials is the MOST probable cause of the occupational illness?

1) Petroleum products and greases
2) Solvents
3) Metals and metal plating
4) Rubber and its compounds

Ans.1 Petroleum products and greases (19%) was more than double that of solvents (8%), triple that of metals and metal plating (6%), and more than 5 times that of rubber and its compounds (3%).

1640. Workers in certain jobs are subjected to a greater risk of contacting occupational dermatitis. Which industry accounts for the highest percentage of cases?

1) Oil refining
2) Synthetic resin manufacturing
3) Automotive manufacture
4) Tanneries

Ans.2 The yearly prevalence of occupational dermatitis for the synthetic resin manufacturing industry (8%) was double its closest runner ups, tanneries (4%) and oil refining (3%). Of the 12 industries surveyed, the automobile manufacturing industry had the least number of cases (0.4%).

1641. The occupation most likely to encounter metal fume fever is:

1) Electroplater
2) Welder
3) Firefighter
4) Miner

Ans.2 Metal fume fever is an acute condition of short duration caused by a brief high exposure to the freshly generated fumes of metals, such as zinc and magnesium or their oxides. Symptoms appear from four to twelve hours following exposure and consist of fever and shaking chills.

1642. Cumulative trauma disorders (CTD's) are rapidly becoming a major source of workers' compensation losses. The most prevalent form of CTD is:

1) Bursitis
2) Carpal tunnel syndrome
3) Kaolinosis
4) Tendonitis

Ans.2 Carpal tunnel syndrome is a common disease of workers performing repeated exertions or movements of the fingers/hand and wrist which compress the median nerve in the carpal tunnel. It is often associated with tingling, pain or numbness in the thumb and first three fingers, frequently resulting in lost time and workers' compensation costs.

1643. Disorders that result from repeated exertions or movements of the body are most often described by which of the following terms?

1) Regional musculoskeletal disorders
2) Cumulative trauma disorders
3) Occupational cervicobrachial disorders
4) Occasional trauma disorder

Ans.2 Disorders that are caused, precipitated, or aggravated by repeated exertion or movements of the body are most frequently called cumulative trauma disorders (CTD's). Other terms less commonly used include: repetitive trauma disorders, repetitive strain injuries, overuse syndromes, and regional musculoskeletal disorders.

1644. Which of the following is LEAST likely to involve problems with occupational dermatitis?

1) Leather tanning
2) Lumbering
3) Asbestos processing
4) Farm labor

Ans.3 The incidence rates reported by OSHA indicate that occupational dermatitis occurs more frequently in leather tanning > farm labor > forest services > asbestos processing.

1645. You have conducted a JSA in the patient-accessible areas of a major hospital. Which of the following occupational hazards was a major concern?

1) Welding machine
2) Hot slag
3) Radionuclides
4) Inert environments

Ans.3 Knowing that the JSA was at a hospital, the most commonly found hazard from this list was probably radionuclides used for radiation therapy and research.

1646. What is the WBGT given a wet bulb temperature of 75ºF and a globe temperature of 89ºF (no solar load)?

1) 85ºF
2) 92ºF
3) 78.5ºF
4) 79.2ºF

Ans.4 WBGT = 0.7 WB + 0.3 GT = (0.7 * 75ºF) + (0.3 * 89ºF) = 79.2ºF
where WB = natural wet bulb temperature obtained with wetted sensor exposed to natural air movement. GT is the globe temperature take from the center of a 6 in. diameter hollow copper sphere painted on the outside with black matte finish (called a globe thermometer).

1647. Six workers are exposed to methylene chloride at the following levels (based on an 8-hr TWA): 27 ppm, 31 ppm,12 ppm, 86 ppm, 27 ppm and 40 ppm. What is their statistical median exposure?

1) 37 ppm
2) 31 ppm
3) 27 ppm
4) 29 ppm

Ans.4 The median number in a set is the middle number, if the number (n) in the set is an odd number. When there are an even number of values in a sequence, the median is found by taking the average of the 2 "middle" numbers, in this case: (31 ppm + 27 ppm) / 2 = 29 ppm

1648. An employee's exposure to a mixture of solvents is found to be 20 ppm to solvent A, 10 ppm to solvent B, and 14 ppm to solvent C. If the toxic effects of these solvents are additive, and their TLV's are 100 ppm, 50 ppm, and 100 ppm respectively, what is the employee's exposure as a percent of the TLV?

1) 44%
2) 54%
3) 64%
4) 74%

Ans.2 To get the total exposure determine the percentage of the TLV of each substance the employee is exposed to and add them together or:
Exposure = (20 ppm/100 ppm)+(10 ppm/50 ppm)+(14 ppm/100 ppm) = 20% + 20% + 14% = 54%

1649. Calculate a worker's daily 8 hr TWA exposure to a solvent with a TLV of 100 ppm if she spends 1 hr at an operation where the concentration is 250 ppm, 4 hr at 200 ppm, and 3 hr at 100 ppm.

1) 69 ppm
2) 135 ppm
3) 169 ppm
4) 1,350 ppm

Ans.3 TWA = [(1 hr * 250 ppm)+(4 hr * 200 ppm)+(3 hr * 100 ppm)] / (1 hr + 4 hr + 3 hr) = 169 ppm

1650. Compute the TWA concentration from the following data taken during the day shift of a worker.
6:00 am - 9:00 am ... 100 ppm

9:00 am - 11:00 am ...200 ppm
11:00 am - 12:00 pm .... 0 ppm
12:00 pm - 2:00 pm ...100 ppm

1) 105 ppm
2) 109 ppm
3) 111 ppm
4) 113 ppm

Ans.4 TWA = $(C_1T_1 + C_2T_2 + C_3T_3 + C_4T_4)/(T_1+T_2+T_3+T_4)$
where C_1=concentration during T_1
T_1 = Time at concentration C_1
C_2 = Concentration during T_2
T_2 = Time at concentration C_2, etc...
TWA = [(100 ppm * 3 hr) + (200 ppm *2 hr) + (0 ppm * 1 hr) + 100 ppm * 2 hr)] / (3 hr + 2 hr + 1 hr + 2 hr) = 112.5 ppm

1651. NIOSH has established accuracy requirements for detector tubes requiring that at one half of the exposure limit the tube will be +/- _____ % and +/- _____ % at one to five times the exposure limit.

1) 25, 35
2) 35, 25
3) 10, 25
4) 25, 25

Ans.2 The NIOSH criteria for detector tubes is that they be accurate to +/- 35% at one half the exposure limit and +/- 25% at one to five times the exposure limit.

1652. Some materials are not collected well on charcoal sorbent tubes. Polar molecules tend to be collected better on:

1) Coconut charcoal tubes
2) PVC
3) Silica gel tubes
4) Both 2 and 3

Ans.3 Silica gel tubes collect polar materials better than charcoal tubes. Materials such as methanol and most amines are collected much better on silica gel. One reminder, water is polar and can be collected in large quantities and may interfere.

1653. The TLV's for gases and vapors are estimated on what basis?

1) Weight to weight
2) Volume to volume
3) Volume to weight
4) Weight to specific density

Ans.2 The unit "ppm" means parts of contaminant per million parts of air (for gases and vapors). This is a volume to volume measurement.

1654. The chemical component of the smoke tubes used in most ventilation studies is:

1) Vanadium pentoxide
2) Titanium tetrachloride
3) Ammonium chloride
4) Dioctylphthalate

Ans.2 Titanium tetrachloride decomposes in moist air to form titanium dioxide and hydrochloric acid. The titanium dioxide is white.

1655. The preferred method of sampling for benzene is:

1) Fritted bubbler with NaOH
2) Activated charcoal tube
3) Portable LIRA
4) Midget impinger with H_2O_2

Ans.2 Other methods have been recommended but the OSHA method is activated charcoal.

1656. To determine an employee's exposure to noise, all noise from ___ dB to 130 dB is accumulated.

1) 80 dB
2) 85 dB
3) 90 dB
4) 95 dB

Ans.1 All continuous, intermittent, and impulse sound shall be integrated into the noise measurement beginning at 80 dB and continuing to 130 dB. 29 CFR 1910.95 (d)(2)(i)

1657. What agent is generally used to desorb a sample from an activated carbon tube?

1) Chloroform
2) Carbon disulfide
3) Toluene
4) Methyl chloroform

Ans2. The most common desorbent is carbon disulfide.

1658. What instrument would be used to check for a refrigerant?

1) DU flame spectrophotometer
2) Halide meter
3) Combustible gas indicator
4) Gas chromatograph

Ans.2 Refrigerants are mainly halogenated hydrocarbons, therefore, the Halide meter is the best instrument.

1659. What is the OSHA method for sampling for trichloroethylene?

1) 100 liters on silica gel

2) 10 liters on hopcalite
3) 10 liters on activated charcoal
4) 10 liter mylar bag

Ans.3 Trichloroethylene, as is true for many organic solvents, is collected on a charcoal tube with a recommended sample size of 10 liters.

1660. What is the best adsorbent for most organic vapors that have boiling points above 0ºC?

1) Charcoal
2) Impinger
3) Filter
4) Silica

Ans.1 The non-polar characteristic of charcoal allows for the adsorbance of organic vapors in preference to atmospheric moisture.

1661. What is the primary OSHA method for sampling for carbon monoxide?

1) Hopcalite direct reading meter
2) Gas detection tube
3) Ecolyzer
4) Activated charcoal tube

Ans.3 The Ecolyzer is the recommended method for sampling for carbon monoxide. It is also recommended that detector tubes be used throughout the day to sample from the worker's breathing zone.

1662. What is the secondary (NIOSH) method for field carbon monoxide determination?

1) Hopcalite catalytic combustion
2) Detector tubes

3) Gas chromatography
4) Infrared absorption spectrometry

Ans.2 Detector tubes that meet the standards of quality in the detector tube certification program are the preferred secondary method of sampling for carbon monoxide in the field.

1663. When measuring for an oxygen deficient atmosphere one could use any of the following EXCEPT:

1) Galvanic sensing cell
2) Paramagnetic gas analyzer
3) Coulometric detector
4) Detector tube

Ans.4 All the listed techniques are commonly available for measuring oxygen concentration. However, a detector tube has an accuracy of +/- 25% and would be inappropriate for measuring oxygen deficiency. The minimum acceptable concentration of oxygen is 19.5% for general industry.

1664. When using a compound such as isoamyl acetate to test the fit of a respirator, one of the limitations is that the individual being fitted:

1) Is at an increased risk of cancer
2) May have olfactory fatigue
3) May not give an honest response
4) Both 2 and 3

Ans.4 For fitting of the respirator to be effective, the person being fitted must detect the odor and tell the tester. This requires that the sense of smell not be fatigued and that the person give an honest answer.

1665. When using a passive monitor to measure employee exposure to organic solvents, monitoring begins:

1) When the employee begins the operation
2) When the cover is removed from the badge
3) When the pump is turned on
4) Before and after lunch break

Ans.2 Monitoring begins when the cover is removed from the badge. Diffusion of the materials begins immediately upon removal of the cover. The time to record is the time the cover was removed.

1666. Which type of filters are used for asbestos counting?

1) Whatman Ashless
2) PVC
3) Mixed cellulose ester
4) Glass with PVC backing

Ans.3 The mixed cellulose ester filter is used because its index of diffraction is the same as the oil that is used in the oil immersion process for fiber counting. The oil and paper disappear, leaving the fibers to be counted.

1667. You determined that the concentration of a material is100 mg/m^3 by using a pump calibrated to 1.0 lpm. You later find out that the pump actually ran at 1.5 lpm. What is the corrected concentration?

1) 66.7 mg/m^3
2) 1.667 mg/m^3
3) 166.7 mg/m^3
4) 6.67 mg/m^3

Ans.1 We actually drew in more air than we thought. We drew in 1.5 * the air we thought. 1 / 1.5 is the ratio of the air sample sizes.
1 / 1.5 = 0.667 0.667 * 100 mg/m^3 = 66.7 mg/m^3

1668. Conversion of TLV's in mg/m^3 to ppm is found in the following formula:

1) = (TLV in mg/m^3) (22.45) / (Gram Molecular Weight of Substance)
2) = (TLV in ppm)(24.45) / (Gram Molecular Weight of Substance)
3) = (TLV in mg/m^3)(24.45) / (Gram Molecular Weight of Substance)
4) = (Gram Molecular Weight of Substance) / (TLV in mg/m^3)(24.45)

Ans.3 Since ppm is a volume/volume ratio, the mass must cancel out, leaving volume units. 24.45 is used for the molar volume since temperature is assumed to be 25ºC.

1669. Instantaneous or grab air samples are used to determine:

1) Composition of the air at a specific point in time and location
2) Average composition of the air during a given time period
3) Number of samples to be taken over an 8-hr shift
4) Compliance with a PEL

Ans.1 Grab or instantaneous samples are used to determine the composition of air at a specific point in time and at a specific location. They are used in many situations including determining concentrations during short exposures and in emergency response situations.

1670. A certain environment contains trichloroethylene, acetone, and xylene. Which of the following field instruments would most likely yield immediate information regarding concentrations of the individual vapors?

1) Combustible gas meter
2) Mass spectrometer
3) Colorimetric detector tubes
4) Passive dosimeters

Ans.3 Colorimetric detector tubes are a field instrument that will yield immediate information regarding the concentrations of the individual vapors of trichloroethylene, acetone, and xylene.

1671. Colorimetric detector tubes may be considered completely adequate for:

1) Evaluation of eight hour TWA's for any of the organic vapors
2) Collection of STEL organic vapor samples
3) Precise determinations by anyone capable of following the manufacturer's instructions
4) Rough determinations when used by someone skilled in their use

Ans.4 Detector tubes results yield rough estimates of instantaneous exposure levels when used by a trained professional. TWA's and STEL's cannot be determined using detector tubes, since they are instantaneous samples.

1672. When evaluating where to collect samples, which of the following should be considered first?

1) Those at risk due to own activity
2) Those nearby
3) Those who exhibit symptoms of exposure
4) Those who complain

Ans.3 The samples should be collected on a range of people. However, those exhibiting symptoms are most at risk and should be evaluated first; followed by those at risk but without symptoms, those who complain, and then others nearby.

1673. A chiller in the plant is suspected to have a leak due to a loss of pressure in the CFC closed loop system. What type of instrument would you use to detect the leak?

1) Detector tube
2) Halide meter
3) LEL meter

4) O_2 meter

Ans.2 Chillers contain refrigerants such as CFC's which are also known as a class of chemicals called halides. Therefore, the best meter would be one that detects halides.

1674. Calculate the TWA of the following exposures:
1 hour = 150 ppm
2 hours = 75 ppm
5 hours = 10 ppm

1) 44 ppm
2) 53 ppm
3) 65 ppm
4) 74 ppm

Ans.1 TWA = [(1 hr * 150 ppm) + (2 hr * 75 ppm) + (5 hr * 10 ppm)] / 8 hr TWA = 43.75 ppm

1675. Of the ways to evaluate the hazards of gases and vapors, all the following are true EXCEPT:

1) May be collected on an open-faced filter and weighed
2) May be collected on filter media
3) May be collected in containers or absorbed
4) May be collected on cloth material

Ans.4 Gases and vapors may not be collected on cloth material.

1676. In evaluating the hazards of fumes and mists, all of the following are true EXCEPT:

1) May be collected on filter media
2) May be absorbed on charcoal
3) May be collected with a cyclone separator
4) May be analyzed using fume arrestors

Ans.4 Fumes and mists may not be analyzed using fume arrestors.

1677. If a worker is exposed to lead for 4 hr at 24 ag/m^3, 3 hr at 48 ag/m^3, and 1 hr at 0 ag/m^3, what is his exposure as an 8-hr TWA?

1) 28 ag/m^3
2) 30 ag/m^3
3) 33 ag/m^3
4) 48 ag/m^3

Ans.2 To get the TWA, you must calculate the exposure rate for each part of the 8-hr day for which there is an exposure.
(24 ag/m^3)(4 hr/8 hr) = 12 ag/m^3
(48 ag/m^3)(3 hr/8 hr) = 18 ag/m^3
(0 ag/m^3)(1 hr/8 hr) = 0 ag/m^3
Then, you need to add together each weighted exposure to get the 8-hr exposure.
TWA = 12 ag/m^3 + 18 ag/m^3 + 0 ag/m^3 = 30 ag/m^3

1678. What is the wet bulb-globe temperature to which an industrial assembly worker is exposed if the following temperatures are obtained:
Dry bulb = 89ºF
Natural wet bulb = 84ºF
Globe = 81ºF

1) 83.9ºF
2) 83.1ºF
3) 84.0ºF
4) 84.7ºF

Ans.2 Use the WBGT formula for indoor applications.
WBGT = 0.7 WB + 0.3 GT,
where WB is the temperature reading from the natural wet bulb and GT is the the globe temperature.

WBGT = (0.7 * 84ºF) + (0.3 * 81ºF)
WBGT = 83.1ºF

1679. What is the natural wet bulb globe temperature to which a construction worker, working outdoors, is exposed if the following temperatures are obtained:
Dry bulb = 92ºF
Natural wet bulb = 88ºF
Globe = 94ºF

1) 89.8ºF
2) 89.6ºF
3) 90ºF
4) 91.3ºF

Ans.2 Use the WBGT formula for outdoor applications.
WBGT = 0.7 WB + 0.2 GT + 0.1 DB, where WB is the temperature from the natural wet bulb, GT is the globe temperature, and DT is the dry bulb temperature.
WBGT = (0.7 * 88ºF) + (0.2 * 94ºF) + (0.1 * 92ºF) WBGT = 89.6ºF

1680. The correct procedure in performing a wipe sample is:

1) Start at center and make increasingly larger circles to edge
2) Start at outside edge and progress to the middle
3) Make perpendicular swipes of area
4) Make parallel swipes of the area

Ans.2 The "OSHA Technical Manual" calls for starting at the edge and making increasingly smaller concentric circles to the middle. The surface area sampled is 100 cm^2.

1681. Assessments for potential TLV-C exposures should be sampled for:

1) 15 min
2) 5 min
3) 1 min
4) Only where instantaneous monitoring is feasible

Ans.1 Where instantaneous monitoring is not feasible, the TLV-C can be assessed by sampling over a 15-min period except for those substances that may cause immediate irritation.

1682. In performing personal air sampling for welding fumes, where should the filter cassette be placed?

1) Attached to the lapel
2) Inside the welding helmet
3) On the shoulder
4) At the waist

Ans.2 The OSHA Technical Manual specifies placement of the filter cassette inside the welding helmet.

1683. In the NIOSH lifting equation, what does RWL represent?

1) Range of weight limits
2) Recommended weight limit
3) Reference work load
4) Recommended work limit

Ans.2 This equation calculates the recommended weight limit for workers. It is used to evaluate two-handed manual lifting tasks.

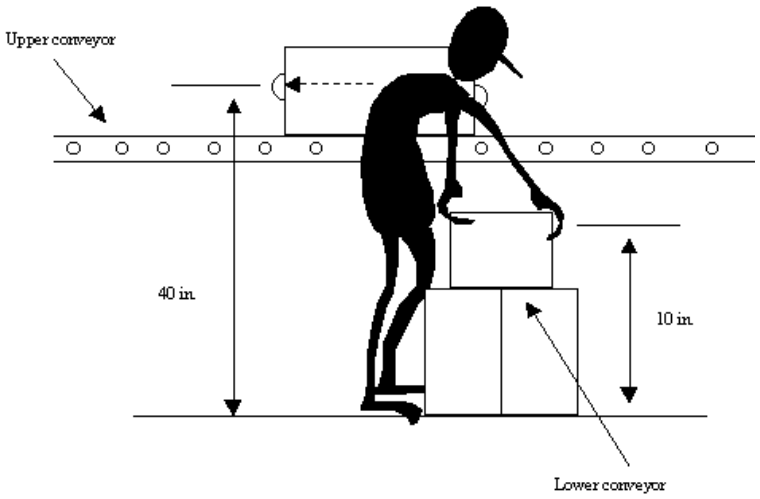
1684. When using the NIOSH lifting equation, you will be able to determine the recommended weight limit for lifting tasks using the formula developed through

research in ergonomics. If the actual load is observed and divided by the recommended weight limit, you will calculate:

1) The optimum task multiplier
2) The severity rate for the task
3) The percent of the population that can do the task
4) The lifting index

Ans.4 The lifting index, or LI , is calculated by dividing the actual load by the recommended weight limit. This is a good way to compare tasks or possible changes to a task.

1685.



You are evaluating a lifting task in your facility that involves moving tote boxes containing parts from a lower conveyor to an upper conveyor at right angles to each other. The tote boxes have two handles and are easy to grasp. The tote box handles, when on the lower conveyor, are 10 in. from the floor. When the tote box is on the upper conveyor, the handles are 40 in. from the floor. The center of the belt of both conveyors is 10 in. from the feet of the worker. One lift is required every minute during the full 8-hr shift. The tote box weighs 4 lb. How much material can be placed in the tote box?

1) 4 lb
2) 12 lb
3) 16 lb
4) 20 lb

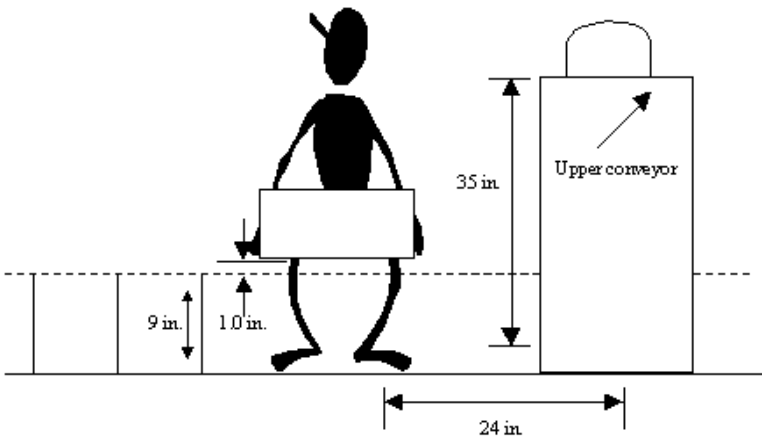
Ans.3 The NIOSH lifting equation is:

RWL = LC * HM * VM * DM * AM * FM * CM
where LC = 51 lb
HM = (10/H)
VM = [1-0.0075 * absolute value of (V-30)]
DM = [0.82 + (1.8/D)]
AM = (1-0.0032 * A)
(All of the above is part of the lifting equation BEFORE we start applying the numbers from the problem.)

These are the numbers we will use in the RWL equation.

H = 10 in. . . . this is the horizontal distance from the feet
V = 10 in. . . . this is the vertical distance from the floor
D = 30 in. . . . this is how far the item is lifted vertically
A = 90º . . . this is the asymmetry (body twisting)
FM = 0.75 ... this is the frequency factor
CM = 1.00 ... this is the coupling factor
RWL = 51 lb(10/10)[1 - (0.0075(absolute 10 - 30)](0.82 + (1.8/30)](1 - (0.0032 * 90)] * 0.75 * 1.00
= 51 * 1.00 * 0.85 * 0.88 * 0.712 * 0.75 * 1.00
= 20.37 lb
Product weight = RWL - tote weight
= 20.37 lb - 4.0 lb
= 16.37 lb

1686.



You are evaluating a lifting task in your facility that involves moving tote boxes containing parts from an upper conveyor to a lower conveyor. The totes are poorly constructed and are difficult to grasp. Most workers simply grasp the bottom of the slippery tote box from the upper conveyor (located 35 in. from the floor) and drop the tote box 1 in. from the lower conveyor (located 9 in. from the floor). The conveyors are at right angles to each other, and the center of the conveyor belts are 24 in. from the feet of the workers. Two lifts are required every minute during a 2-hr shift. The

tote box weighs 4 lb. If the weight of parts placed in the tote box is 8 lb, what is the lifting index? Should changes be made?

1) 1.22/Changes should be made
2) 0.81/Changes are not necessary
3) 0.78/Changes are not necessary
4) 1.15/Changes should be made

Ans.1 The NIOSH lifting equation is:

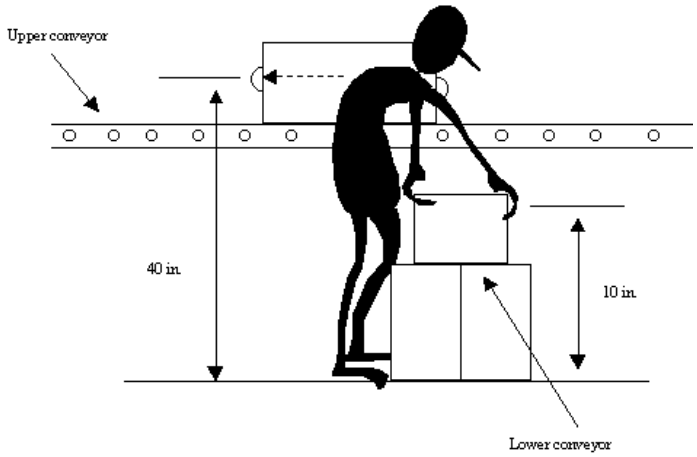
RWL = LC * HM * VM * DM * AM * FM * CM

where LC = 51 lb
HM = (10/H)
VM = [1-0.0075 * absolute value of (V-30)]
DM = [0.82 + (1.8/D)]
AM = (1-0.0032 * A)
(All of the above is part of the lifting equation BEFORE we start applying numbers from the problem.)

These are the numbers we will use in the RWL equation.
H = 24 in. . . . this is the horizontal distance from the feet
V = 35 in. . . . this is the vertical distance from the floor
D = 25 in. . . . this is the lifted distance (the workers drop the tote the last inch - that is why D is not 26 in.)
A = 90º . . . this is the asymmetry (body twisting)
FM = 0.84 ... this is the frequency factor
CM = 0.90 ... this is the coupling factor
RWL = 51 lb(10/24)[1 - (0.0075(absolute 35 -30)] [0.82 + (1.8/25)] [1 -(0.0032 * 90)] * 0.84 * 0.90
= 51 lb * 0.417(0.9625) 0.892 * 0.712 * 0.84 * 0.90 = 9.8 lb
Total load = 4.0 lb + 8.0 lb = 12 lb
Lifting Index = Load / Recommended Weight Limit
= 12 lb / 9.8 lb = 1.2

A lifting index of over 1.00 is too high and indicates that changes to the process and/or to the magnitude of the load should be made to reduce the lifting index to 1.00 or less.

1687.



You are evaluating a lifting task in your facility that involves moving tote boxes containing parts from a lower conveyor to an upper conveyor at right angles to each other. The tote boxes have two handles and are easy to grasp. The tote box handles, when on the lower conveyor, are 10 in. from the floor. When the tote box is on the upper conveyor, the handles are 40 in. from the floor. The center of the belt of both conveyors is 10 in. from the feet of the worker. One lift is required every minute during the 8 hour shift. The tote box weighs 4 lb. If the lower conveyor is raised 10 in., how much more product can be placed in the tote box without exceeding the RWL?

1) 1.3 lb
2) 2.6 lb
3) 3.0 lb
4) 15 lb

Ans.2 The NIOSH lifting equation is:

RWL = LC * HM * VM * DM * AM * FM * CM
where
LC = 51 lb
HM = (10/H)
VM = [1-0.0075 * absolute value of (V-30)]
DM = [0.82 + (1.8/D)]
AM = (1-0.0032 * A)
(All of the above is part of the lifting equation BEFORE we start applying the numbers from the problem.)

These are the numbers we will use in the RWL equation.
H = 10 in. . . . this is the horizontal distance from the feet
V = 10 in. . . . this is the vertical distance from the floor
D = 30 in. . . . this is how far the item is lifted vertically
A = 90º . . . this is the asymmetry (body twisting)
FM = 0.75 ... this is the frequency factor
CM = 1.00 ... this is the coupling factor

RWL_1 = 51 lb(10/10) [1 - (0.0075 absolute 10 - 30)](0.82 + (1.8/30)](1- 0.0032 * 90) * 0.75 * 1.00 = 20.37 lb RWL_1 is the initial state.
When we change things, we must recalculate the new RWL. In this problem, the only things that change are the vertical distances.
V changes to 20 in. from 10 in.
D changes to 20 in. from 30 in.
RWL_2 = 51 lb (10/10)[1-(0.0075 * absolute 20-30)] [0.82 + (1.8/20)] [1-(0.0032 * 90)] * 0.75 * 1.00 = 22.92 lb Difference = 22.92 lb - 20.37 lb = 2.55 lb

1688. Under foundry conditions, the numerical value of the WBGT Index is calculated using the formula:

1) 0.7 WB + 0.3 GT
2) 0.7 WB + 0.2 GT + 0.1 DB
3) 0.7 WB + 0.1 GT + 0.1 DB
4) 0.7 WB + 0.2 GT + wind velocity

Ans.1 The wet bulb globe temperature (WBGT) is calculated indoors or outdoors with no solar loads:
WBGT = 0.7 WB + 0.3 GT
Foundry conditions would exclude the solar load. Had the question asked for the formula outdoors with solar load, the answer would have been:
WBGT = 0.7 WB + 0.2 GT + 0.1 DB where WB = Natural wet bulb temperature, GT = Globe temperature, DB = Dry bulb temperature

1689. How much an employee can safely lift depends least on:

1) Body size
2) Frequency
3) Body position
4) Distance

Ans.1 Although body size can contribute to overall strength, it is not a good predictor of potential injury. When designing a lifting task and safe weight limits, the most important factors are how often the load is lifted (frequency), how far the load is carried (distance), and how the load is lifted (body position).

1690. In the course of an eight-hour work shift, a maintenance mechanic works TWICE in the vicinity of an operation where the measured carbon monoxide concentration in the air at his breathing zone level is fairly constant at 50 ppm. The first time he works for 3 hr in the morning, and the second time he works 2.5 hours in the afternoon. All the other times, he works in areas of the plant where his exposure to carbon monoxide is essentially zero.What is his time-weighted average (TWA) exposure to carbon monoxide?

1) 17.2 ppm
2) 34.4 ppm
3) 51.6 ppm
4) 103.2 ppm

Ans.2 TWA=(3 hr * 50 ppm + 2.5 hr * 50 ppm + 2.5 hr * 0 ppm)/8 hr
= [(150 + 125)hr-ppm] / 8 hr
= 34.4 ppm

1691. A fiber is defined as a particle with an aspect ratio (L:W) of:

1) 2:1
2) 3:1
3) 4:1
4) 5:1

Ans.2 Fibers are particles with an aspect ratio of 3:1, that is, the length of the fiber is three times the width.

1692. The particle size of fumes range from:

1) 0.01 - 0.2 microns
2) 0.001 - 0.02 microns
3) 0.001 - 0.002 microns
4) Fumes are not particles

Ans.1 Fumes are solid particles that are condensed from metal oxides. They range in size from 0.01 - 0.2 microns in diameter. Particles greater than 0.5 microns are considered dusts.

1693. The particle size of mists range from:

1) Mist are liquids-particle size does not apply
2) 0.4 - 4 microns
3) 4 - 40 microns
4) 40 - 400 microns

Ans.4 Mists are a suspension of liquid particles from condensation of vapor. They range in particle size from 40 - 400 microns.

1694. The terminal settling velocity of a particle is:

1) Inversely proportional to its diameter
2) Directly proportional to its diameter
3) Directly proportional to the square of its diameter
4) Inversely proportional to the square of its diameter

Ans.3 The settling velocity of particles is described by Stoke's Law: Terminal Settling Velocity = $0.006(\text{specific gravity})(\text{diameter})^2$. This relationship shows that the settling velocity is proportional to the square of the diameter of the particle.

1695. Smoke tubes can be utilized in which of the following situations:

1) In hot, dry air
2) At velocities below 150 ft/min
3) When exact measurements are required
4) At velocities below 175 ft/min

Ans.2 The smoke tube is not applicable for velocities greater than 150 ft/min because the rapid air flow dissipates the smoke.

1696. When investigating an indoor air quality issue, the first step is to:

1) Take quantitative air samples
2) Inspect the building HVAC system
3) Meet with affected employees and management
4) Survey the area for sources of air contaminants

Ans.3 The first step in investigating an indoor air quality problem is to understand the nature of complaints from the affected employees. This will provide clues as to the potential contaminants or other sources of the problem. Air sampling is premature since potential contaminants are unknown. Inspecting the HVAC system or surveying for sources of contaminants may not be necessary if complaints point in a different direction.

1697. The concentration of carbon dioxide where symptoms of exposure begin to appear is:

1) 350 ppm
2) 600 ppm
3) 1,000 ppm
4) 5,000 ppm

Ans.3 Many studies have shown that hypersensitive employees begin to report headaches, fatigue, dizziness, and other symptoms at about 1,000 ppm. 360 ppm is the concentration of carbon dioxide in a rural environment. Inside a well ventilated building, carbon dioxide is usually about 600 ppm. The ACGIH TLV for carbon dioxide is 5,000 ppm.

1698. Using the equation for adding decibels, what is the sound pressure level if the following 3 noise sources are added together: 80 dB, 85 dB, and 92 dB?

1) 89 dB

2) 93 dB
3) 95 dB
4) 102 dB

Ans.2 Use the formula to add decibels:
$Lpt = 10 \log [10^{(P_1/10)} + 10^{(P_2/10)} + ... 10^{(P_n/10)}]$
$Lpt = 10 \log [10^{(80/10)} + 10^{(85/10)} + 10^{(92/10)}]$
$= 10 \log (2.0011 \times 10^9)$
= 93 dB

1699. If an additional noise source of 120 dB is added to 3 others (80 dB, 85 dB, and 92 dB), what is the resultant sound pressure level?

1) 94 dB
2) 120 dB
3) 125 dB
4) 213 dB

Ans.2 The formula for adding decibels is:
$Lpt = 10 \log [10^{(P_1/10)} + 10^{(P_2/10)} + ... 10^{(P_n/10)}]$
$= 10 \log (1.0020 \times 10^{12})$
= 120 dB

1700. When calculating estimated sound pressure levels in a room, R refers to:

1) Room constant
2) Directivity of the noise
3) Size of the room
4) None of the above

Ans.1 The room constant is an indication of the amount of sound absorption in a room. The smaller the room constant, the more reverberant the room.

1701. If a sound pressure level of 92 dB is obtained at 3 ft from a machine, what level should be expected at 8 ft from the machine, assuming free-field, non-directional noise source?

1) 82.1 dB
2) 83.5 dB
3) 85.4 dB
4) 89.0 dB

Ans.2 Use the equation for calculating the effect of distance on sound pressure:
dB0 = dB1 + 20 log (d1/d0) where db0 is the pressure at the source, db1 is the pressure at distance 1, d0 is the distance where db0 is measured, and d1 is the distance where db1 is measured
92 = dB1 + 20 [log (8/3)]
92 = dB1 + 20 * 0.4260
dB1 = 83.5

1702. The maximum exposure to impulsive or impact noise should not exceed:

1) 90 dBA peak
2) 115 dB peak
3) 120 dB peak
4) 140 dB peak

Ans.4 Exposure to impulsive or impact noise should not exceed 140 dB peak sound pressure level. 29 CFR 1910.95 (b)(1)

1703. Of the following, which sound measuring instrument is the LEAST commonly used for most noise problems encountered in a manufacturing plant?

1) Sound level meter
2) Octave-band analyzer
3) Noise dosimeter
4) Oscilloscope

Ans.4 In a manufacturing plant, with a variety of noise problems, noise measuring equipment most commonly used include sound level meters, octave-band analyzers, and noise dosimeters.
SOUND LEVEL METER - This is the basic direct reading instrument used to measure sound pressure variations in air. The frequency range is usually 20 to 20,000 Hz.
OCTAVE-BAND ANALYZER - This instrument is used to determine where the noise energy lies in the frequency spectrum. It is an essential piece of equipment for engineering control of noise producing machines or equipment.

NOISE DOSIMETER - This instrument is worn by a worker to accumulate a record of the noise energy to which he/she was exposed during the work shift.

1704. A 73 dB sound pressure level (SPL) was measured at 600 ft. What is the SPL at 150 ft?

1) 75 dB
2) 78 dB
3) 85 dB
4) 90 dB

Ans.3 For each halving of the distance you must add 6 dB. There were two halvings of the distance; therefore, add 12 dB.
Answer = 85 dB

1705. An enclosure is placed around a worker and a noise source. Why is the sound pressure level experienced by the worker greater than without the enclosure?

1) Reverberation
2) Impact
3) Masking
4) Increase in line component

Ans.1 Enclosing a noise source always increases the sound pressure level because direct sound is added to the reflected sound.

1706. An octave band analyzer is an instrument used to determine:

1) The noise frequency distribution
2) SPL in A, B, and C weighted scales
3) Peak to peak SPL's
4) A sound level within a fraction of a decibel

Ans.1 The octave band analyzer input microphone divides the sound into frequency bands that are called octaves. It measures the sound pressure levels in each band. These data are often essential for designing effective controls.

1707. During noise level surveys the microphone should be:

1) Pointed at the source
2) Perpendicular to the source
3) Does not matter what orientation
4) Is determined by the manufacturer

Ans.4 Instruments are different and the operating manual must be reviewed before using to ensure correct operation.

1708. How often is calibration of sound level meters required?

1) Monthly
2) Quarterly
3) Before each survey
4) Before and after each survey

Ans.4 Good practice dictates that the meter be calibrated before and after each survey.

1709. Which instrument is required for assessing an individual's hearing ability?

1) Sound level meter
2) Octave band analyzer
3) Audiometric booth
4) Noise dosimeter

Ans.3 The sound level meter and octave band analyzer are used for area sound measurements. The noise dosimeter is used for making noise dose measurements in the individual's hearing zone. The only way to assess hearing is with an audiometric booth.

1710. When might you use the C-weighting on a sound level meter?

1) For estimating adequacy of hearing protective devices
2) For measuring very high frequency noise
3) For measuring community noise from aircraft
4) When trying to most closely imitate the human ear response

Ans.1 The C-weighting on a sound level meter can be used to estimate the actual reduction in noise level provided by hearing protection. The estimate is performed in three steps:
Step 1 - Monitor the noise level in the workplace using the C-weighted scale
Step 2 - Convert the measured noise level to a C-weighted time-weighted average (TWA)
Step 3 - Subtract the noise reduction rating (NRR) specified by the haring protection manufacturer from the C-weighted TWA found in Step 2.
The value found in Step 3 is based on the C-weighted scale and can be reliably used as a conservative estimate of the benefit of wearing hearing protection in high noise environments.

1711. Which of the following is used for detecting alpha radiation?

1) Film badge
2) Thermoluminescence detectors (TLD)
3) Pocket dosimeter
4) Geiger-Mueller with very thin windows

Ans.4 A Geiger-Mueller detector is used for radiation survey measurements because it is capable of detecting very small amounts of radiation. With very thin windows in the probes, the instrument can be used to detect alpha radiation. Without them, the instrument is used to detect beta, gamma, and x-rays. Note that this instrument does not give a uniform response for different radiation energy levels and is accurate only for the type of radiation energy for which it is calibrated.

1712. Which of these meter(s) would be used most often for monitoring weekly exposure to x-rays?

1) Film badge
2) Thermoluminescence detectors (TLD)
3) Pocket dosimeter
4) Both 1 and 2

Ans.4 The film badge and thermoluminescence detectors would be used most often for monitoring weekly exposure to x-rays. The film badge, worn on the outer clothing, will allow an estimate of an accumulated dose of radiation. Thermoluminescence detectors are dosimeters worn as a body badge or finger ring. Most commonly, they are small chips of lithium fluoride with an affinity for storing energy (accumulated dose of x-rays) that can be released on a readout using a TLD readout instrument. The pocket dosimeter is a direct reading instrument shaped like a pen. It is the instrument of choice when an individual finds it necessary to determine the radiation dose at any time he or she is working.

1713. The amount of material undergoing radioactive decay is often measured in:

1) Rem
2) Rad
3) Curie
4) Roentgen

Ans.3 A curie is the amount of a material that will undergo 3.7×10^{10} disintegrations per second.

1714. How are the readings from a shake-out sample best reported?

1) Mppcf
2) mg/m^3 total
3) mg/m^3 respirable dust
4) % silica

Ans.3 Shake-out refers to the shake-out operation in a foundry. The principal hazard is that of silica dust, hence the best answer is mg/m^3 respirable dust.

1715. IDLH means:

1) Industrial department, labor and health
2) Immediately dangerous to life or health
3) Identification, detection, and limitations
4) Nothing pertaining to safety

Ans.2 This is a term which NIOSH defines as an atmosphere in which staying longer than 30 minutes will cause irreversible effects or hinder the workers ability to exit.

1716. Infrared analysis of breathing air is generally acceptable unless _____ contamination is a concern.

1) CO
2) CO_2
3) HC
4) Mercury

Ans.4 CO, CO_2, and hydrocarbons all absorb strongly in the infrared region of the electromagnetic spectrum. Mercury does not.

1717. Of the following airborne contaminants, which is likely to have the smallest average particle size?

1) Zinc fumes
2) Graphite dusts
3) Silica dusts
4) Hydrocarbon mists

Ans.1 Condensation aerosols such as metal fumes tend to have the smallest particle sizes.

1718. One of the shortcomings of the taste test for qualitatively fitting a respirator is that there is:

1) A risk of harm to the person being fitted
2) Wide variation in the ability to taste saccharin
3) The smoke is very irritating
4) Wide variation in the ability to taste banana oil

Ans.2 One of the shortcomings of the taste test is that there is a wide variation in the ability to taste saccharin. This implies different levels of protection for different people.

1719. Passive monitors have been increasing in popularity in recent years. The monitors depend on two methods of collection. The method used most commonly for organic vapors is _____.

1) Active diffusion
2) Passive diffusion
3) Diffusion according to Fick's Law
4) Both 2 and 3

Ans.4 Organic vapors are allowed to passively diffuse across a membrane into some collection device. One common collection device is activated charcoal. Fick's Law describes the mathematics of passive diffusion.

1720. Personal portable samplers are better than fixed samplers in determining:

1) Average concentrations
2) Peak concentrations
3) Sources of emission
4) Reduction of emission

Ans.1 There may be some overlap in meanings of the terms; however, in common usage "average" is taken to mean TWA. Therefore the personal samplers are best for determining the average exposure of a worker.

1721. ACGIH Biological Exposure Indices (BEI's) help to measure employees' exposure to all of the following, EXCEPT:

1) Biohazards
2) Occupational benzene exposure
3) Non-occupational benzene exposure
4) Hazardous gas exposure

Ans.1 Biohazards are potentially infectious agents of biological origin, which are not addressed by the biological exposure indices. BEI's are the medical sampling determinants for a person's exposure to occupational and non-occupational hazardous chemicals. BEI's are published by ACGIH.

1722. Which of the following cannot be monitored using a cholinesterase test?

1) DDT
2) DVP
3) Parathion
4) Malathion

Ans.1 DDT is the only chemical in the above group that is not a cholinesterase inhibitor.

1723. The odor of diethylamine resembles:

1) Garlic
2) Dead fish
3) Ammonia
4) Xylene

Ans.3 Diethylamine has a decidedly ammonia- like odor.

1724. What instrument would be best to check for leaks of a CFC refrigerant?

1) Mass spectrometer
2) Halide meter
3) Combustible gas indicator
4) Gas chromatograph

Ans.2 Although a mass spectrometer or a gas chromatograph can detect chlorofluorocarbons (CFC's), a more specific instrument such as a halide meter can more easily and quickly detect the presence of halogens (chlorine and fluorine) from the CFC vapor. A combustible gas indicator cannot be used since CFC's are not combustible.

1725. Speech interference levels can be determined with which of the following instruments?

1) Sound level meters

2) Impact noise analyzers
3) Octave band analyzers
4) Noise dosimeters

Ans.3 Speech occurs in the 500 - 2,000 Hz range. Identifying the noise levels in this frequency range requires an octave band analyzer.

1726. The best method for monitoring a long haul truck driver's exposure to noise is through the use of:

1) Sound level meter
2) Octave band analyzer
3) Noise dosimeter
4) Annual audiogram

Ans.3 Sound level meters and octave band analyzers require constant readings. Annual audiograms are a measure of the effects of noise on hearing. Noise dosimeters measure dose and are perfect for this application.

1727. The wet bulb, dry bulb, and psychrometric charts are used together to determine :

1) Water temperature
2) Ambient temperature
3) Radiant heat
4) Relative humidity

Ans.4 Using the wet bulb thermometer and the dry bulb thermometer, one can estimate the relative humidity using a psychrometric chart.

1728. The formula for calculating WBGT with solar load is:

1) WBGT = 0.7 WB + 0.3 GT
2) WBGT = 0.3 WB + 0.7 GT
3) WBGT = 0.1 WB + 0.2 GT + 0.7 DB

4) WBGT = 0.7 WB + 0.2 GT + 0.1 DB

Ans.4 The formula for calculating WBGT with solar load is WBGT = 0.7 WB + 0.2 GT + 0.1 DB

1729. Dry bulb temperature is read from:

1) Psychrometric table
2) Psychrometer
3) Thermometer
4) Dry bulb device

Ans.3 Dry bulb temperature is read from a thermometer.

1730. If an employee is exposed to greater than 85 dBA on average for the eight hour workday, that worker must be included in a hearing conservation program and be:

1) Provided with hearing protection
2) Trained and retrained semi-annually
3) Given an audiogram within 6 months
4) Both 1 and 3

Ans.4 Under OSHA's Hearing Conservation standard, when an employee is exposed to 85 dBA or more over an 8-hour TWA, he/she must be included in a Hearing Conservation Program, provided hearing protection, and given a baseline audiogram (in most cases, the audiogram is required within 6 months). Each employee must be retrained on an annual basis. 29 CFR 1910.95 (c)(1) 29 CFR 1910.95 (g)(1)

1731. When using a sound level meter for measuring an intermittent source, the meter should be set to:

1) Fast response, A-weighting
2) Slow response, no weighting
3) Slow response, A-weighting
4) Intermittent response, flat scale

Ans.3 Measurements of continuous or intermittent noise shall be measured at slow meter response, A-weighting.
29 CFR 1910.95 (a)

1732. Employers must monitor noise levels for employees exposed to more than:

1) 90 dBA
2) 85 dBA
3) 80 dBA
4) 75 dBA

Ans.2 The threshold for employee monitoring is 85 dBA. 29 CFR 1910.95 (d)(1)

1733. How many 150 dB impacts per day is permissible?

1) 50
2) 100
3) 1,000
4) 0

Ans.4 Impact or impulse noise should never exceed 140 dBA. 29 CFR 1910.95 (b)(1)

1734. A charcoal tube sample was taken by drawing 40 liters of air through the tube which was then analyzed for benzene. If 0.5 mg of benzene was found on the tube, what was the concentration of benzene in the air sample assuming STP? (MW of benzene is 78)

1) 3.92 ppm
2) 4.10 ppm
3) 3.50 ppm
4) 4.05 ppm

Ans.1 Since 1 m^3 = 1,000 liters
40 liters = 0.040 m^3
0.5 mg/0.040 m^3 = 12.5 mg/m^3
Next convert mg/m^3 to ppm.

ppm = [(mg/m^3) 24.45] / MW
ppm = (12.5 mg/m^3 * 24.45) / 78
= 3.92 ppm

1735. Hopcalite may be used in the detection of which of the following?

1) NO_2
2) CO
3) CO or CO_2
4) NO_2 or CO_2

Ans.2 Hopcalite is commonly used in the detection of carbon monoxide monitors.

1736. What would be the TLV of a mixture containing 35% by weight methylchloroform (TLV = 1,900 mg/m^3), 35% chloroform (TLV = 50 mg/m^3), and 30% perchloroethylene (TLV = 170 mg/m^3)?

1) 50 mg/m^3
2) 112 mg/m^3
3) 96 mg/m^3
4) 50 ppm

Ans.2 TLV(mixture) = 1/Σ(f/TLV)
Where f = fraction of specific contaminant
TLV = TLV for the specific contaminant
TLV(mixture) = 1/[(0.35/1,900 + 0.35/50 + 0.3/170)mg/m^3]
TLV(mixture) = 1/0.0089 mg/m^3
TLV(mixture) = 112 mg/m^3

1737. A length of stain sulfur dioxide indicator tube indicates a breathing zone concentration of 15 ppm +/- 25% at barometric pressure of 650 mm Hg. What is the corrected concentration?

1) 17.5 ppm +/- 25%
2) 15.0 ppm +/- 25%

3) 14.5 ppm +/- 25%
4) 13 ppm +/- 25%

Ans.1 To correct for barometric pressure: 15 ppm (760 mm Hg/650 mm Hg) = 17.5 ppm

1738. When used outdoors with solar radiation present, the WBGT requires measurement of all of the following EXCEPT:

1) Dry bulb temperature
2) Globe temperature
3) Natural wet bulb temperature
4) Wet globe/dry bulb temperature

Ans.4 When solar radiation is present, the wet bulb globe temperature requires measurement of the dry bulb temperature, globe temperature, and natural wet bulb temperature.

1739. For indoor use, when solar radiation is not present, the WBGT requires:

1) Globe temperature and natural wet bulb temperature
2) Dry bulb temperature and globe temperature
3) Dry bulb temperature and natural Wet bulb temperature
4) Globe temperature, natural Wet bulb temperature, and dry bulb temperature

Ans.1 When solar radiation is not present, the wet bulb globe temperature requires measurement of the globe temperature and natural wet bulb temperature.

1740. In the evaluation of heat stress, the natural wet bulb globe temperature combines the effect of all of the following EXCEPT:

1) Natural wet bulb temperature includes the effect of air temperature, humidity, and air velocity
2) Dry bulb temperature includes the effect of air temperature
3) Globe temperature includes the effect of radiant heat
4) Bott's Ball

Ans.4 Air temperature, humidity, air velocity and radiant heat all need to be considered when measuring heat stress. The natural wet bulb temperature, dry bulb temperature, and globe temperature consider all of these factors.

1741. The A-weighted sound level measurement is the preferred scale of measure because:

1) It distinguishes intermittent and impact noise
2) Weighting is related to effects of noise on the ear
3) It filters high frequency background noise
4) It can assess temperature and pressure

Ans.2 The A-weighting most closely weights the sound to the injurious effects of the noise on the ear.

1742. The origin of the "A" weighting network on a sound level meter is to simulate the response of the human ear to all of the following EXCEPT:

1) Relatively low levels of sound
2) At 40 phons
3) At sound pressure levels less than 55 dB
4) 1,000 Hz

Ans.4 "C" weighting is at 1,000 Hz.

1743. The sum of two sound intensities of 70 dB each is approximately:

1) 71 dB
2) 73 dB
3) 76 dB
4) 140 dB

Ans.2 Doubling the sound intensity results in a three decibel increase.

1744. A cholinesterase test is used on workers exposed to:

1) Trichloroethylene
2) Parathion
3) Mercury
4) Lead

Ans.2 Parathion is a cholinesterase inhibitor. A test for cholinesterase will help to determine exposure to parathion. The other answers are not cholinesterase inhibitors.

1745. A wet bulb thermometer is part of the equipment used to determine:

1) Air temperature
2) Effective temperature
3) Relative humidity
4) Radiant heat load

Ans.3 The wet and dry bulb thermometer together may be used with a psychrometric chart or table to determine absolute and relative humidity.

1746. A spirometer is used to measure:

1) Stack velocities
2) Air flow from sampling equipment
3) Exhaust volumes
4) Lung respiratory volumes

Ans.4 A spirometer is used to measure lung respiratory volumes to determine lung function. A pitot tube with a manometer or a velometer is used to measure stack velocities and exhaust volumes. Air flow from sampling equipment is measured by a rotameter.

1747. Which of the following types of radiation is not detected by film badges?

1) Alpha

2) Beta
3) Gamma
4) X-rays

Ans.1 Only alpha radiation is not detected by current film badges.

1748. The coefficient of variation of an asbestos sampling technique has been estimated to be 0.22. The coefficient of variation for charcoal tube sampling has been estimated to be 0.10. This means asbestos air sampling is:

1) More variable
2) Less variable
3) Does not relate to variability of the data
4) No decision can be made based on the data given

Ans.1 The coefficient of variation is the standard deviation divided by the true value multiplied by 100. This means that the higher the coefficient of variation, the more variability there is in the data.

1749. What does TWA mean?

1) Total weighted average
2) Total weighted attitude
3) Time weighted average
4) Time without answers

Ans.3 TWA is the acronym for Time Weighted Average. This is expressed in exposure for an eight hour day. Therefore, if an exposure was the same and continuous for eight hours, it would be the same as the exposure. If the exposure lasted for only four hours, the TWA would be divided by 2 to get the 8-hour exposure.

1750. Which of the following is an example of biological monitoring?

A. Pulmonary Function Test
B. Audiometry
C. CBC
D. Urinalysis

1) A and B
2) A, C, and D
3) C and D
4) All of the above

Ans.3 Biological monitoring is the analysis of blood, urine, skin, etc., to determine the presence of agents or metabolites. Pulmonary function testing and audiometry are biological function measurements; they are examples of medical monitoring.

1751. From a statistical standpoint grab sampling is _____ continuous sampling when trying to verify compliance with an OSHA 8-hour TWA standard.

1) Simpler than
2) More difficult than
3) About as difficult as
4) Treated in exactly the same way as

Ans.2 Verifying compliance with an OSHA TWA using grab sampling is extremely difficult. The preferred method is to take a full shift sample.

1752. A TLV posted in the "Notice of Intended Changes" is considered for adoption after how many years if no evidence comes to light that questions the appropriateness of the value?

1) 1 year
2) 2 years
3) 3 years
4) The time limit is set by the head of OSHA

Ans.1 A TLV posted in the "Notice of Intended Changes" is considered for adoption after 1 year, if no evidence comes to light that questions the appropriateness of the value.

1753. The critical organ for the soluble form of radon is the:

1) Bone
2) Thyroid
3) Lung
4) Spleen

Ans.3 The lung is the organ most impacted by both the soluble and the insoluble forms of radon. Lung cancers can arise from elevated radon exposures.

1754. What is the critical organ for the soluble form of radium?

1) Bone
2) GI tract
3) Lung
4) Spleen

Ans.1 The soluble form of radium is incorporated into the bone, leading to bone cancers. The critical organ for the insoluble form of radium is the GI tract. Other critical organs for the soluble form of radionuclides include:

Isotope	Critical Organ
Tritium	Total body
Strontium	Bone
Iodine	Thyroid
Plutonium	Bone

1755. The ACGIH has established cold stress TLV's for workers in cold occupations. At what body core temperature (ºF) is the worker in danger of developing severe hypothermia if his/her core temperature dropped further?

1) 75.5ºF
2) 80.4ºF
3) 85.8ºF
4) 91.4ºF

Ans.4 Once the core body temperature falls below 91.4ºF, severe hypothermia sets in rapidly. Normally the oral temperature is 98.6ºF and rectal temperature 99.6ºF.

1756. "In vivo" toxicology refers to studies conducted with the following methodology.

1) Toxic agent is added to tissue slices or tissue homogenate
2) Toxic agent is administered to live animals
3) Toxic agent is modeled mathematically
4) Toxic agent is administered to human volunteers

Ans.2 The most correct answer is that "in vivo" toxicology refers to toxicity studies conducted with live animals who are eventually sacrificed and the effect measured. "In vitro" toxicology refers to toxicity studies conducted upon tissue slices or homogenate.

1757. Toxicity recognizes several different mechanisms of multiple chemical interaction. What is potentiation?

1) A substance which does not have a toxic effect upon a system or organ, but when in combination with another chemical, makes that chemical much more toxic
2) Two substances, which when together, interfere with each other's actions
3) The combined effect of these two chemicals is much greater than the sum effect of each agent acting independently
4) Both 2 and 3

Ans.1 Potentiation is a substance which does not have a toxic effect upon a system or organ, but when in combination with another chemical, makes that chemical much more toxic. A classic example of this effect is isopropanol and carbon tetrachloride. Isoproponal is not toxic to the liver but when it is added to carbon tetrachloride, the toxicity of the CCl_4 to the liver is greater.

1758. A classic example of synergism, regarding a toxic effect, involves asbestos coupled with a synergistic agent to produce lung cancer. What is this synergistic agent?

1) Water
2) Polyvinyl chloride
3) Dust
4) Cigarette smoke

Ans.4 The classic example of toxicologic synergy is asbestos and cigarette smoke.

1758. A classic example of synergism, regarding a toxic effect, involves asbestos coupled with a synergistic agent to produce lung cancer. What is this synergistic agent?

1) Water
2) Polyvinyl chloride
3) Dust
4) Cigarette smoke

Ans.4 The classic example of toxicologic synergy is asbestos and cigarette smoke.

1759. Relative humidity is calculated using:

1) Wet bulb and dry bulb temperatures
2) WBGT and GBT
3) Dry bulb to GBT
4) Wet bulb and dry bulb temperatures with a psychrometric chart

Ans.4 To calculate relative humidity, first determine the wet bulb and dry bulb temperatures, then use a psychrometric chart to determine the actual relative humidity.

1760. Ionizing radiation has the greatest effect on which tissues or organs?

1) Kidneys
2) Liver
3) Blood
4) Bone marrow

Ans.4 Damage to the blood can be overcome if there is replacement of the injured blood cells by new ones from the bone marrow. However, radiation damage can injure the bone marrow so that cell replacement cannot take place. Damage to the body will be permanent in such cases. Kidneys and the liver are generally highly resistant to ionizing radiation and will be injured only by very high dosages such as those which result from the presence of an internal source.

1761. The effects of ionizing radiation on this body tissue or organ is generally temporary after exposure stops.

1) Skin
2) Eyes
3) Nervous system
4) Hair

Ans.4 Radiation will lead to loss of hair. This effect is generally temporary after exposure stops.

1762. What is the length of time permissible for an exposure to noise of 94 dB?

1) 1.6 hrs
2) 2.3 hrs
3) 4.0 hrs
4) 4.6 hrs

Ans.4 Use the formula for calculating the time weighted average for exposure:
$T = 8 / 2^{[(L-90)/5]}$
where L is measured exposure, and 90 is the permissible exposure
$T = 8 / 2^{[4/5]}$
$= 8 / 1.74$
$= 4.6$ hr

1763. For employees with a significant threshold shift, hearing protector attenuation must be sufficient to reduce exposure to a time-weighted average of:

1) 75 dB
2) 80 dB
3) 85 dB
4) 90 dB

Ans.3 For employees with a significant threshold shift, hearing protector attenuation must be sufficient to reduce exposure to a time-weighted average of 85 dB. 29 CFR 1910.95 (i)(2)

1764. The correct term for hearing loss that can be induced by occupational exposure to noise is:

1) Presbycusis
2) Sociocusis
3) Sensorineural
4) Conductive

Ans.3 Presbycusis is hearing loss due to age. Sociocusis is loss due to background noise. Conductive loss is a loss between the outer ear and the cochlea.

1765. The damaging effect of noise on hearing depends on:

1) The sound pressure level of the noise
2) The frequency distribution of the noise
3) The transmission/frequency lapse rate
4) Both 1 and 2

Ans.4 The sound pressure level and the frequency distribution of the noise along with the length of exposure will determine the effect on the ear. The transmission/frequency lapse rate is fictitious.

1766. When incurring a Temporary Threshold Shift due to a noisy workplace, most of the TTS occurs:

1) Right before the end of the exposure period
2) In the first few hours of exposure
3) Towards the end of the first week of exposure
4) Occurs within minutes of entering the area

Ans.2 Most of the TTS incurred is incurred within the first 1-2 hours of the exposure.

1767. What type of ionizing radiation is the greatest external hazard?

1) Alpha
2) Beta
3) Gamma
4) Delta

Ans.3 Gamma rays are the most penetrating of all radiation and have the ability to penetrate the skin.

1768. The range of a beta particle is any of the following EXCEPT:

1) Finite
2) Varies from zero to a maximum
3) Depends on the radionuclide
4) Does not depend on transmission medium

Ans.4 Beta radiation is not monoenergetic as is an alpha emission. A particular beta particle can have a range from zero to a maximum depending on the particular radionuclide. The range of particles IS dependent on the medium of transmission.

1769. What does an "alpha" emission consist of?

1) An electron
2) An electromagnetic wave
3) 2 neutrons and 2 protons
4) 1 neutron and 1 proton

Ans.3 An "alpha" emission consists of 2 neutrons and 2 protons (a helium nucleus). This type of ionizing radiation is easily shielded so it is considered an internal radiation hazard as it can cause heavy damage in tissue.

1770. In the alpha decay of radium to radon, the following occurs: ${}^{226}_{88}RA \rightarrow$ alpha particle + ${}^{x}_{z}RN$
What is the value of "X" in this decay equation?

1) 222
2) 223
3) 224
4) 226

Ans.1 An alpha particle is a helium nucleus. It contains 2 neutrons and 2 protons. Losing 4 atomic mass units converts radium 226 to radon 222.

1771. In the alpha decay of radium to radon, the following occurs: ${}^{226}_{88}RA \rightarrow$ alpha particle + ${}^{x}_{z}RN$. What is the value of "Z" in this decay equation?

1) 85
2) 86
3) 87
4) 88

Ans.2 An alpha particle contains 2 neutrons and 2 protons. Therefore, the release of one alpha particle decreases the element's atomic number by 2 (protons are counted for the atomic number).

1772. Which of the following forms of ionizing radiation are considered electromagnetic?

1) Beta
2) Gamma
3) X-ray
4) Both 2 and 3

Ans.4 Beta and alpha radiation is considered particulate, as sub-atomic particles are involved. X-ray and gamma radiation are electromagnetic radiation.

1773. What does MeV stand for?

1) 10^1 electron volts
2) 10^3 electron volts
3) 10^6 electron volts
4) 10^9 electron volts

Ans.3 Some people confuse this nomenclature with KeV. 1 KeV is 10^3 electron volts, and 1 MeV is 10^6 electron volts.

1774. In the beta decay of strontium to yttrium, the following occurs: $^{90}_{38}Sr$ → beta particle + $^{x}_{39}Y$. What is the value of "X" in this decay equation?

1) 89
2) 90
3) 91
4) 92

Ans.2 The "X" refers to the mass number for yttrium. As beta decay causes no change in the mass number, the mass number remains 90 for yttrium.

1775. In the beta decay of strontium to yttrium, the following occurs: $^{90}_{38}Sr$ → beta particle + $^{x}_{39}Y$. What is the value of "Z" in this decay equation?

1) 37
2) 38
3) 39
4) 42

Ans.3 Beta decay causes an increase of +1 in the atomic number as a negative charge is leaving the atom. A +1 increase in the atomic number increases the atom's ranking on the periodic table to the next higher element. Therefore "Z" is 39.

1776. Recently, health physicists have been switching from curies to becquerels. A curie is 3.7×10^{10} disintegrations per second (dps). What is a becquerel (Bq)?

1) 1.0×10^{10} dps
2) 3.7×10^{10} dps
3) 0.5×10^{10} dps
4) 1 dps

Ans.4 Becquerels (Bq) are based upon the relationship that 1 Bq is 1 dps. A curie is equivalent to 3.7×10^{10} Bq.

1777. What is a roentgen?

1) The amount of X-ray or gamma radiation that produces ionization resulting in 1 electrostatic unit of change in 1 cm^3 of dry air at STP
2) The absorption of 100 ergs of energy from any radiation in 1 g of any material
3) The "biological" dose to living tissue
4) The same as a rem

Ans.1 A roentgen is a measurement of exposure whose definition is the amount of X-ray or gamma radiation that produces ionization resulting in 1 electrostatic unit of change in 1 cm^3 of dry air at STP. The measurement of an absorbed dose, or a rad, is defined as the absorption of 100 ergs of energy from any radiation in 1 g of any material. A rem is a biological dose.

1778. Which of the following is not a symptom of carpal tunnel syndrome?

1) Numbness
2) Loss of feeling in grip
3) Loss of hand use and function
4) All of the above are symptoms

Ans.4 Loss of hand use and function can occur in the most extreme cases. Carpal tunnel syndrome is caused by serious degradation of the median nerve. Phalen in the 1960's described a condition called "ape hand" caused by atrophy of the large muscle of the thumb.

1779. When the wrist is bent toward the thumb side of the hand, this is known as a(n)_____ deviation.

1) Ulnar
2) Radial
3) Palmar
4) Dorsi

Ans.2 Ulnar deviation is a bend toward the little finger. A palmar flexion is a bend toward the palm, and a dorsiflextion is an extension of the wrist towards the back part of the hand.

1780. TLV's (Threshold Limit Values) are based on all of the following EXCEPT:

1) Industrial experience
2) Human studies
3) Animal studies
4) Referrals

Ans.4 TLV's are based on the best available information from industrial experience, from human and animal studies, and when possible, from a combination of the three.

1781. When analyzing air sampling concentration data, the data are assumed to be _____ distributed.

1) Exponentially
2) Normally
3) Gaussian
4) Log normally

Ans.4 Air sampling data are assumed to be log normally distributed. If there is some question about the distribution one should perform one of a variety of statistical tests to determine which distribution the data fits best.

1782. To convert rads to rems, a multiplier is used that quantifies the radiation damage to biological systems based upon the specific radiation in question. What is the multiplier called?

1) Quality factor
2) Conversion factor
3) Roentgen equivalent man
4) X factor

Ans.1 The multiplier used to convert rads to rems is termed the quality factor or QF.
QF (X, gamma, beta) = 1
QF (neutrons of unknown strength) = 10
QF (alpha) = 20
The relationship is: 1 rem = dose in rads * QF

Nuclear Regulatory Commission, Code of Federal Regulations, Title 10, 20.1004

1783. What do asbestos, silicon dioxide, metal fumes, nitrogen dioxide, and beryllium have in common?

1) They are carcinogens
2) They are lung damaging agents
3) They are nephrotoxins
4) They are nerve damaging agents

Ans.2 Asbestos, silicon dioxide, metal fumes (iron, cadmium, zinc oxides) and beryllium are lung damaging agents. In order for a harmful agent to exert its toxic effect, it must come into contact with a body cell. To make that contact, entry into the body is made as a result of the inhalation of the foregoing in the form of dusts, fibers, or particles which are deposited and retained in the lungs, where they (depending on dose response) have the potential to produce signs, symptoms, and findings of pulmonary disease. (Asbestos and beryllium are also carcinogenic.)

1784. What fundamental concept in toxicology is used to determine the safety or hazard of chemical substances?

1) Assessment of the hazard and its impact on the population

2) Defining safety and health limits of chemical agents
3) Establishing that a relationship exists between the dose of an agent and the response that it produced in a biologic system
4) Identifying mechanisms by which noxious substances gain access to the biological system

Ans.3 Toxicology is the study of noxious effects of chemical and physical agents. The most fundamental concept in toxicology states that a relationship exists between the dose of an agent and the response that it produces in a biologic system. The first toxicity data on an uncharacterized agent are usually obtained using test animals. A range of doses is found where at the upper end all animals die, and at the lower end, they survive.

1785. At a given intensity, sound in which frequency is most likely to result in permanent hearing damage?

1) 37.5 - 500 Hz
2) 1,000 - 4,000 Hz
3) 8,000 - 16,000 Hz
4) 16,000 - 32,000 Hz

Ans.2 The ear is most susceptible to noise in the range that it is most sensitive. The ear is most sensitive in the range 1,000 - 4,000 Hz.

1786. At what frequency does noise induced hearing loss first show up on an audiogram?

1) 250 - 1,000 Hz
2) 3,000 - 6,000 Hz
3) 250 - 500 Hz
4) 250 - 8,000 Hz

Ans.2 The first sign of noise induced hearing loss is usually seen in the higher frequencies.

1787. The main part of the body involved in maintaining the body's heat balance is:

1) Skin
2) Heart
3) Lungs
4) Hair

Ans.1 The skin is responsible for dissipating heat load through the blood flow of the capillaries. The capillaries expand and contract depending on the load.

1788. The middle ear contains a group of bones called the:

1) Stapes
2) Ossicles
3) Incus
4) Eustachian tubes

Ans.2 The middle ear consists of the eustachian tubes, the mastoid air cell system, and the sound conducting mechanism composed of the incus, malleus, and the stapes. These three small bones are called the ossicles.

1789. The part of the ear which contains the hair-like cells that detect sounds is the:

1) Semicircular canals
2) Cochlea
3) Stapes
4) Anvil

Ans.2 The cochlea is the organ which contains these hair cells. The sound waves move the hairs at specific locations in the cochlea. The sites of hair movement correspond to the sounds' frequencies.

1790. Which frequency of vibration most adversely affects humans?

1) 1 - 125 Hz
2) 140 - 250 Hz
3) 275 - 500 Hz
4) 500 - 1,000 Hz

Ans.1 The 1 - 125 Hz band has been shown to be the most effective in affecting humans.

1791. What size of particles are removed by the mouth and nasal passages?

1) Greater than 1 micron
2) Greater than 5 microns
3) Greater than 10 microns
4) Greater than 20 microns

Ans.3 The mouth and nasal passages can remove particles greater than 10 microns in diameter from impaction on the wet surfaces of the nose and mouth.

1792. Deposition of liquid particles in the respiratory system is dependent on:

1) Particle size
2) Specific gravity
3) Viscosity
4) Solubility in water

Ans.4 A liquid's solubility in water and its reactivity are the determining factors in the deposition of liquid particles in the respiratory system, as compared with solid particles, which depend on specific gravity and particle diameter.

1793. If an employee works in an area for 2.5 hours, what is his/her maximum permissible exposure to noise?

1) 96 dB

2) 97 dB
3) 98 dB
4) 100 dB

Ans.3 Using $T = 8 / 2^{[(L-90)/5]}$ where L is the measured exposure, 90 is the permissible exposure, and T is the exposure time
$T = 8 / 2^{[(L-90)/5]}$
$2.5 = 8/x$
$x = 3.2$
Substituting, $3.2 = 2^{[(L-90)/5]}$ or $3.2 = 2^{y}$
$y = 1.59$
Substituting,
$1.59 = (L-90)/5$
$L = 98$ dB

1794. The average person contains about 0.1 microcuries of K-40 (half life = 1.27×10^6 years). How many dps is this?

1) 3.7×10^4 dps
2) 3.7×10^3 dps
3) 4.6×10^6 dps
4) 2.6×10^3 dps

Ans.2 By definition 1 microcurie = 3.7×10^4 dps; therefore, 0.1 microcurie = 3.7×10^3 dps

1795. What material should be used to provide shielding against gamma rays?

1) Aluminum sheet
2) Paper
3) Pyrex
4) Lead

Ans.4 Gamma rays must be shielded by dense materials such as lead. The thicker and denser the material the better.

1796. A gamma radiation source reads 500 milliroentgens/hr at a distance of 1 m. How far away must a worker be so that the level does not exceed 25 milliroentgens/hr?

1) 3.25 m
2) 4.47 m
3) 5.42 m
4) 6.23 m

Ans.2 $I_2 = (I_1 D_1^2) / D_2^2$
I = Intensity and D = Distance
25 mRoent/hr = [500 mRoent/hr $(1 m)^2$] / D^2
25 mRoent/hr * D^2 = 500 m^2-mRoent/hr
D = 4.47 m

1797. Bone marrow and the eyes are very sensitive to radiation. Damage to the body may be permanent. What other tissue or organ is most sensitive to radiation?

1) Skin
2) Hair
3) Blood
4) Reproductive organs

Ans.3 White blood cells are the most sensitive to radiation, which reduces their number, leaving the body open to infection. Skin is easily damaged; however, the skin has great recuperative powers. Radiation will lead to loss of hair. This effect is generally temporary after exposure stops; however, the new hair may be of a new color or have other characteristics different from the original. To produce sterility in a person would require almost a fatal dose. Genetic effects produced by radiation to reproductive cells can only be surmised, and would not be known for several generations after exposure.

1798. Which type of ionizing radiation is penetrating; whole-body exposure?

1) Alpha
2) Beta
3) Gamma
4) Microwave

Ans.3 Gamma - Penetrating; whole-body exposure
Beta - Primarily skin exposure
Alpha - Will not penetrate dead cells on surface of skin
Microwave - This is a form of non-ionizing radiation

1799. Which type of ionizing radiation is primarily a skin penetrating exposure hazard?

1) Alpha
2) Beta
3) Gamma
4) Infrared

Ans.2 Beta - Primarily skin exposure
Gamma - Penetrating; whole-body exposure
Alpha - Will not penetrate dead cells on surface of skin
Infrared - This is a form of non-ionizing radiation

1800. Which type of ionizing radiation will not penetrate the dead skin cells on the surface of the skin?

1) Alpha
2) Beta
3) Gamma
4) Ultraviolet

Ans.1 Alpha - Will not penetrate dead cells on surface of skin
Beta - Primarily skin exposure
Gamma - Penetrating, whole-body exposure
Ultraviolet - This is a form of non-ionizing radiation

1801. Alpha emitters are:

1) Noted for long half lives
2) External hazards
3) Internal hazards
4) Non-hazardous

Ans.3 Alpha emitters are internal hazards.

1802. What is a beta particle?

1) Helium nucleus component with a double charge
2) High-speed electron
3) X-rays from a nucleus
4) Unknown radiation component in a calibration

Ans.2 A beta particle is usually a high speed electron created in isobaric decay. If the high speed electron is positively charged the particle is a positron.

1803. Electric fields of a high voltage may accelerate electrons to a high enough speed whereby they can escape their orbitals and induce X-rays. What is the magnitude of this high voltage?

1) Greater than 10 kV
2) Greater than 5 kV
3) Greater than 2 kV
4) Greater than 1 kV

Ans.1 Electric fields of greater than 10 kV can cause ionizing radiation to be generated.

1804. What is the maximal whole body lifetime dose of ionizing radiation to a 50 year old U.S. worker allowable under current OSHA and NRC guidelines?

1) 5 rem
2) 10 rem
3) 100 rem
4) 160 rem

Ans.4 Utilizing the total dose accumulation equation:
Total Dose Accumulation = 5 (N-18) rem; where N = age
= 5 (50-18)
= 5 * 32

= 160 rem
29 CFR 1910.1096 (b)(2)(ii)

1805. One of the laws governing ideal gas behavior that industrial hygienists must account for when conducting air sampling is Charles' Law. This law describes the relationship of:

1) Volume with pressure and temperature
2) Pressure with noise level and temperature
3) Temperature with pressure and enthalpy
4) Volume with absolute temperature

Ans.4 Charles' Law states that the volume of a gas sample varies directly with absolute temperature (e.g., temperature in Kelvin or Rankine). To determine the volume of an ideal gas at a new temperature, use Charles' Law.
New Volume (V_2) = Starting Volume (V_1)(T_2/T_1)
$V_2 = V_1 (T_2/T_1)$

1806. Which of the following diseases would be least likely classified as an occupational hazard?

1) Yellow fever
2) Q fever
3) Brucellosis
4) Ringworm

Ans.1 Pet shop salesmen, stockmen, and breeders of cats and dogs are occupations that may encounter ringworm. Slaughterhouse workers, farmers, and veterinary surgeons are occupations having potential to be infected with Q fever, a rickettsial organism. Workers with cattle and pigs, beef and pork handlers in slaughterhouses, and farmers can contract brucellosis (undulant fever). Yellow fever is caused by a mosquito bite.

1807. Which of the following is the least likely occupational disease?

1) Asbestosis
2) Emphysema
3) Siderosis

4) Glanders

Ans.2 Asbestosis and siderosis are a kind of pneumoconiosis involving specific lung changes caused by the inhalation of dust or fumes. In the past, asbestos was widely used in construction materials (roofing, siding, wallboard, floor tiles, insulation); textiles (fireproof clothing, blankets); shipbuilding materials (pipe insulation); and automotive parts (brake shoes and clutch pads). Excessive fumes (iron oxide) from welding operations may produce siderosis with a pigmentation of the lungs without disability (black in welders lungs and red in iron ore miners). Glanders is a bacterial disease that may be contracted by horse trainers, veterinary surgeons or anyone having contact with sick horses or mules. Emphysema is most commonly associated with cigarette smoking.

1808. At 10ºC and 4.0 atm, a sample of gas occupies 1,000 liters. If the gas is reduced in volume to 300 liters and the temperature remains constant, what will the pressure be?

1) 10 atm
2) 13 atm
3) 14 atm
4) 23 atm

Ans.2 Boyle's law states that at constant temperature, pressure and volume are inversely related - as the pressure increases the volume decreases.
P (atm) = 4.0 atm(1,000 liters / 300 liters)
= 4.0 atm (3.333)
= 13.33 atm

1809. The atomic number of an element indicates:

1) The total number of neutrons
2) The total number of protons
3) The total number of protons and neutrons
4) The total number of protons minus neutrons

Ans.2 The atomic number refers to the total number of protons in the nucleus.

1810. Bonds formed between two different atoms often result in the electrons not being shared equally between them. This unequal sharing is a function of how _____ one atom is, relative to the other.

1) Polar
2) Covalent
3) Electronegative
4) Energetic

Ans.3 Electronegativity is the property associated with the unequal sharing of electrons during bonding. One atom has a greater ability to attract electrons. When the bond is formed, the electrons spend more time near one atom. Large differences in electronegativity will cause a transfer of electrons (i.e., an ionic bond).

1811. Carbon monoxide is formed by:

1) Degradation of carbon dioxide
2) High temperature combustion greater than 700ºC
3) Incomplete combustion
4) Chlorofluorocarbon interaction

Ans.3 Carbon monoxide is an intermediate oxidation product of carbon. If enough oxygen were present the carbon would have been oxidized to carbon dioxide. When there is insufficient oxygen to fully oxidize the carbon to carbon dioxide, carbon monoxide is formed.

1812. Concentrations of gases and vapors are commonly expressed as:

1) Parts per million by weight
2) Parts per million by volume
3) Millions of particles per cubic foot (MPPCF)
4) Particles per million

Ans.2 Gases and vapors are not particles and therefore cannot be measured in units of MPPCF or "particles per million."

1813. The least likely outcome of exposure to asbestos fibers is:

1) Asbestosis
2) Skin cancer
3) Mesothelioma
4) Lung cancer

Ans.2 Asbestos is a generic term applied to a number of hydrated mineral silicates, including chrysotile, amosite, crocidolite, tremolite, and anthophyllite. Asbestos causes asbestosis, cancer of the lungs and digestive tract, and mesothelioma. Asbestosis and asbestos cancer are the long term effects of inhaling asbestos fibers (they do not become clinically evident until more than 20 years have passed from onset of exposure) whereas, mesotheliomas result from a short term exposure and are diffuse and spread rapidly. Death usually results within one year of diagnosis. Skin cancer is not a likely outcome of asbestos exposure.

1814. Which of the following chemical asphyxiants inhibits cytochrome oxidase and is rapidly fatal?

1) Carbon monoxide
2) Hydrogen sulfide
3) Hydrogen cyanide
4) Both 2 and 3

Ans.4 Both hydrogen cyanide and hydrogen sulfide inhibit the enzyme cytochrome oxidase, which is essential for tissue respiration. Tissue hypoxia and death may occur rapidly. Carbon monoxide interacts with the hemoglobin molecule to produce asphyxia.

1815. Which of the following chemicals is NOT a hepatotoxic agent?

1) Carbon tetrachloride
2) Nitrosoamines
3) Tetrachloroethane
4) Arsine

Ans.4 Arsine is a gas whose primary effect upon inhalation is hemolysis (break down of red blood cells). All of the other chemicals are severe hepatotoxic agents. Carbon tetrachloride produces severe diffuse central necrosis of the liver, tetrachloroethane produces yellow atrophy of the liver, and nitrosoamines and aflatoxins cause severe liver damage.

1816. Insoluble uranium is well known for its radioactive risks. What organ system is potently impacted by its non-radioactive, toxic properties?

1) Eyes
2) Kidneys
3) Liver
4) Olfactory

Ans.2 Insoluble and soluble uranium are potent kidney toxins which also can affect the skin, lungs, bone marrow, and lymphatics.

1817. Which chemical is characteristically related to oat cell carcinoma of the lung?

1) Asbestos
2) Vinyl chloride
3) Chromium
4) Bis-chloromethyl ether

Ans.4 Prolonged exposure to bis-chloromethyl ether has been shown to cause oat cell carcinoma of the lung. The other characteristic cancers are listed below:
Asbestos - Mesothelioma
Vinyl chloride - Angiosarcoma
Nickel, wood dust, chromium - Adenocarcinoma of the nasal cavity

1818. What was the characteristic cancer exhibited by the radium dial painters earlier in this century?

1) Osteogenic sarcoma
2) Mesothelioma
3) Angiosarcoma
4) Oat cell carcinoma of the lung

Ans.1 Radium dial painters were afflicted with osteogenic sarcoma from radium exposure. Osteogenic means "originating in bone." Sarcoma means "a malignant neoplasm arising in connective tissue and especially in bone, muscle, or cartilage."

1819. Thalidomide is a classic example of what type of agent?

1) Mutagenic
2) Teratogenic
3) Toxic
4) Radioactive

Ans.2 The two potential answers were "mutagenic" and "teratogenic". Thalidomide is a classic teratogen. Thalidomide babies were born without limbs because the drug's effect acted on the developing fetus. Teratogens have an adverse effect upon the developing fetus. Mutagens modify the actual genetic material.

1820. Benzene is metabolized by the body to form which metabolite?

1) Urea
2) Phenol
3) Cyclohexane
4) Hexane

Ans.2 Phenol is the classic metabolic product of benzene. Total urinary phenols is the method of choice for benzene exposure biological monitoring.

1821. Methylene chloride is a mild toxicant whose primary metabolite is:

1) Phenol
2) Carbon monoxide
3) Methyl chloroform
4) Benzene

Ans.2 Methylene chloride is metabolized by the body to form carbon monoxide. Due to this fact, combined exposure to this solvent and carbon monoxide must be taken into account and is considered by both the ACGIH and NIOSH safe exposure level recommendations.

1822. A physician that specializes in hearing disorders is an:

1) Audiologist
2) Audiothamologist
3) Otolaryngologist
4) Ear, nose, and throat specialist

Ans.3 An audiologist conducts and reads audiograms. An ear, nose, and throat specialist will diagnose hearing problems, but does not specialize in hearing disorders as an otolaryngologist does.

1823. Cumulative trauma disorder (CTD) is:

1) An acute injury caused by repetitive motion
2) A chronic illness caused by exposure to extremes in temperature
3) A chronic injury caused by an accumulation of micro-injuries
4) An acute form of carpal tunnel syndrome

Ans.3 Cumulative trauma disorder is a type of chronic musculoskeletal injury that is caused by an accumulation of many micro-injuries over a period of time. Acute or macro-injuries cause immediate pain and damage from sudden or extreme force, while micro-injuries are from low-intensity stresses that accumulate over a long period of time.

1824. Which of the following is NOT a cumulative trauma disorder?

1) Thoracic Outlet Syndrome
2) Tenosynovitis
3) Raynaud's Phenomenon
4) De Quervain's Disease

Ans.3 Raynaud's Phenomenon (or Syndrome) is a disorder of the hand and arm sometimes resulting from excessive vibration exposure, but not as a result of cumulative trauma. Raynaud's Phenonemon can develop from the single use of a vibrating tool, though it typically comes from long-term use. In addition, it can be caused by exposure to certain chemicals, exposure to cold, and excessive emotional stress. While doctors can diagnose and treat the signs and symptoms of Raynaud's Phenonemon, they are still not sure what causes it. Thoracic Outlet Syndrome is due to repetitive compression of nerves and blood vessels in the neck and shoulder. Tenosynovitis is due to repetitive compression of nerves and blood vessels in the wrist. De Quervain's Disease is due to repetitive compression of nerves and blood vessels in the thumb.

1825. The primary factors that affect development of cumulative trauma disorder (CTD) include all of the following EXCEPT the:

1) Age of the worker
2) Intensity of force applied
3) Frequency of repetition of high risk tasks between high risk tasks
4) Lack of a proper rest period

Ans.1 The age of a worker has little effect on the development of CTD. The accumulation of micro-injuries from excessive force is compounded by the frequency of repetition of such high risk tasks. (This takes into account the lack of rest between tasks).

1826. Which of the following is NOT a symptom of carpal tunnel syndrome?

1) Cannot tell hot from cold
2) Excessive sweating of hand and wrist
3) Lack of strength
4) Trouble performing simple manipulative tasks

Ans.2 Carpal tunnel syndrome patients often develop autonomic nerve damage which results in loss of the sweat function, leaving dry, shiny skin. The remainder of the answers are common symptoms of carpal tunnel syndrome.

1827. The decibel is the commonly used unit for expressing sound pressure. Decibels must be calculated as compared to a reference level. The reference level for sound pressure is 2×10^{-5} newtons per square meter (or 2×10^{-5} pascals). Why was this level selected?

1) It is the lowest sound pressure which can be measured
2) It approximates the threshold of hearing
3) It simplifies the calculations
4) It approximates the pain threshold

Ans.2 The reference level for sound pressure of 2×10^{-5} was chosen because it approximates the threshold of human hearing at 1,000 Hz.

1828. A sound pressure level of 3 N/m^2 corresponds to how many decibels?

1) 101.3 dB
2) 78.6 dB
3) 103.5 dB
4) 85.9 dB

Ans.3 Lp = 20log (P/Po), where
Lp = pressure level, P = the measured pressure, and Po is the reference sound pressure of 2×10^{-5} Pa
Lp = 20log (3/0.00002)
= 20log (150,000)
= 103.5 dB

1829. If the sound pressure of 3 N/m^2 is reduced in half to 1.5 N/m^2, what is the new sound pressure level in decibels?

1) 100.5 dB
2) 51.75 dB
3) 85.5 dB
4) 97.5 dB

Ans.4 Use the formula Lp = 20log (P/Po) where Lp is the pressure level, P is the measured pressure and Po is the reference pressure of 2×10^{-5} Pa
Lp = 20log (1.5/0.00002)
= 20log (75,000)
= 20 * 4.8751
= 97.5 dB

1830. Sound power is used to describe:

1) The sound source
2) Amount of acoustical energy produced over time
3) The amplitude of a sound source normalized to a frequency of 10,000 Hz
4) Both 2 and 3

Ans2. Sound power is used to describe the acoustical energy produced over time from a sound source.

1831. A unit used to describe sound power is:

1) Watts
2) Joules
3) Ergs
4) Lamberts

Ans.1 Sound power is measured in watts.

1832. To express sound power (in watts) as decibels, what is required?

1) This is not possible
2) A sound level meter
3) A reference level
4) Knowledge of the sound intensity

Ans.3 Decibels require a reference level. For sound power, the reference level is 10^{-12} W.

1833. What is the sound power in decibels of a machine with a sound power of 1,000 W?

1) 100 dB
2) 120 dB
3) 140 dB

4) 150 dB

Ans.4 Using the sound power formula, Lp = 10 log I/Io where I is the measured pressure and Io is the reference pressure of 10^{-12} Pa
Lp = 10 log ($1{,}000/10^{-12}$)
= 10 log (1×10^{15})
= 150 db

1834. A machine normally produces 1,000 watts of sound power. What is the sound power (in decibels) of the machine if its sound power is reduced by a factor of 10?

1) 15 dB
2) 140 dB
3) 100 dB
4) 135 db

Ans.2 One thousand (1,000) watts reduced by a factor of 10 is 100 watts.
Use the formula for sound power:
Lp = 10 log I/Io db where I is the measured pressure and Io is 10^{-12} Pa
Lp = 10 log ($100/10^{-12}$)
= 10 log (1×10^{14})
= 140 db

1835. Sound intensity is defined as:

1) The loudest octave band
2) Total acoustic power
3) The sound power per unit area
4) The sound pressure level actually heard

Ans.3 Sound intensity is defined as the sound power per unit area.

1836. A unit used to describe sound intensity is:

1) Watts per square meter
2) Joules per second
3) Ergs per liter
4) Feet per second

Ans.1 Sound intensity is described by watts (power) per square meter (area).

1837. The agent responsible for paralytic shellfish poisoning is:

1) Curare
2) Saxitoxin
3) Tetrodotoxin
4) Ciguateratoxin

Ans.2 Saxitoxin is actually a complex of toxins produced by the Gonyaulax species dinoflagellates (kingdom Protista). Humans develop paralytic shellfish poisoning as a consequence of ingesting filter feeding mollusks which have accumulated the dino-flagellates. Paralytic shellfish poisoning is associated with algal blooms.

1838. _____ is the neurotoxin found in the liver of the pufferfish.

1) Tetrodotoxin
2) Saxitoxin
3) Batrachotoxin
4) Curare

Ans.1 In pufferfish poisoning, death is due to skeletal muscle paralysis. Both saxitoxin and tetrodotoxin have applications in mechanistic toxicity studies because they selectively block the sodium channels along the axons, preventing the inward sodium current while leaving the outward potassium current unaffected.

1839. Botulinum toxin interferes with the propagation of nerve impulses by preventing the release of _____ from the axon terminal.

1) Acetylcholine
2) Norepinephrine
3) Adrenaline
4) Gaba

Ans.1 Botulinum toxin, produced by the bacterium clostridium botulinum, binds irreversibly to the axon terminal preventing the development of the end plate potential on the muscle side of the synapse. The muscle behaves as if it were denervated. The result is the paralysis characteristic of botulism.

1840. The principle effects of sulfur dioxide on uncompromised human subjects are:

1) Carcinogenic effects
2) Irritant properties
3) Teratogenic potential
4) Pulmonary toxicity

Ans.2 Sulfur dioxide (SO_2), a major form of air pollution, causes reflex bronchoconstriction and increases in resistance to pulmonary flow. These are mainly a consequence of irritant properties. Atmospheric particulates (e.g., particulate sulfates) can enhance the effects of SO_2 on the body.

1841. Most snake venoms are:

1) Complex mixtures
2) Rapidly fatal at small doses
3) Neurotoxic
4) Produced by pit vipers

Ans.1 Snake venoms are complex mixtures containing numerous proteins as well as small amounts of metals, glycoproteins, lipids, biogenic amines, and inorganics such as sodium, calcium, and potassium. Many of the proteins have enzymatic activity (e.g., hyaluronidase, collagenase, proteolytic enzymes) which are thought to facilitate the venoms toxicity.

1842. Sound can be transmitted through all of the following EXCEPT:

1) Water
2) Steel
3) Lead
4) Vacuums

Ans.4 Sound can be transmitted through all of the other media. Vacuum is an absence of a medium; therefore, sound cannot be transmitted through it.

1843. Which characteristic of quantifying vibration levels does not take into account the time history of the wave?

1) Velocity
2) Peak to peak
3) Average
4) None of the above

Ans.2 A peak to peak does not take into account the time history of the wave.

1844. What is a worker's daily 8 hour exposure to trichloroethylene if he spends 90 min at 300 ppm, 4.5 hr at 200 ppm, and 2.0 hr at 20 ppm?

1) 17.4 ppm
2) 65.2 ppm
3) 114 ppm
4) 174 ppm

Ans.4 TWA = [(1.5 hr * 300 ppm) + (4.5 hr * 200 ppm) + (2.0 hr * 20 ppm)] / (1.5 hr + 4.5 hr + 2.0 hr) = 174 ppm

1845. Airborne zinc oxide resulting from welding galvanized iron would be classified as:

1) Dust
2) Smoke
3) Fume
4) Mist

Ans.3 Fumes are produced from materials which are solid at room temperature, are vaporized, and then condense. They often are metal oxides. The zinc oxide would therefore be classified as a fume.

1846. The vapor pressure of benzene at 25ºC is 97 mm Hg. What is the maximum concentration of benzene vapor in air that can be obtained at 25ºC at a total pressure of 760 mm Hg?

1) 0.127 ppm
2) 1.27 ppm
3) 1,270 ppm
4) 127,000 ppm

Ans.4 Cppm = (Pvapor / Ptotal) x 10^6 ppm
= (97 mm Hg / 760 mm Hg) x 10^6 ppm
= 127,000 ppm

1847. Why is 1,1,1-trichloroethane sometimes substituted for trichloroethylene?

1) Lower flash point
2) Less volatile
3) Higher boiling point
4) TLV is higher

Ans.4 The TLV is higher for 1,1,1-trichloroethane. Also trichloroethylene is suspected of having carcinogenic activity.
NOTE: These materials are not in frequent use due to the environmental elimination of CFC's from the work place.

1848. Disregarding the exceptions, in a single quarter, what is the maximum allowable whole body occupational exposure to ionizing radiation?

1) 1 rem
2) 1 1/4 rem
3) 1 1/2 rem
4) 2 rem

Ans.2 There are exceptions allowing whole body exposure up to 3 rem per calendar quarter based on the age of the employee, and the type of other exposures the employee has received. 29 CFR 1910.1096 (b)(1)

1849. All of the following standards call for the use of respirators in the event of an over-exposure EXCEPT:

1) Lead
2) Asbestos
3) Benzene
4) Carbon dioxide

Ans.4 Lead, asbestos, and benzene have an OSHA standard associated with their use. In each standard there is mention of respiratory protection. There is no standard for carbon dioxide.

1850. What would be the best way to reduce radiant heat emissions in an industrial furnace?

1) Use shielding
2) Employ blowing cool air
3) Lower the furnace temperature
4) Use modern heat absorption chemicals in the furnace

Ans.1 Radiant heat is best controlled by shielding.

1851. Find the density of carbon disulfide if its specific gravity is 1.26.

1) 1.3 lb/gal
2) 1.6 lb/gal
3) 6.6 lb/gal
4) 10.5 lb/gal

Ans.4 Specific gravity is the ratio of the density of the material to that of water. Since water has a density of 8.33 lb/gal, the carbon disulfide must have a density of 1.26 times that of water.
Density = 1.26 * 8.33 lb/gal
= 10.5 lb/gal

1852. There is a piston with an outside force of 4,000 psi acting on a volume. Keeping all other factors constant, how will the volume change if the piston force is increased to 8,000 psi? Assume that the volume contains an ideal gas.

1) Decreases by a factor of 10
2) Decreases by a factor of 4
3) Decreases by a factor of 2
4) Increases by a factor of 2

Ans.3 Use the ideal gas laws (knowing that T is constant).
$P_1V_1 = P_2V_2$
If the pressure is increased by a factor of 2, the volume must decrease in the same proportion.
$V_2 = (V_1P_1) / P_2$
$= (V_1P_1) / (2\ P_1)$
$= (V_1 / 2)$

1853. 100 mg/m^3 of HCl equals how many ppm at 25ºC and 760 mm Hg?

1) 61 ppm
2) 67 ppm
3) 69 ppm
4) 76 ppm

Ans.2 The molecular weight of HCL is 36.5 (1 chlorine atom at 35.5 plus 1 hydrogen atom at 1).

PPM = (mg/m^3 * 24.45) / MW PPM
= 100 mg/m^3 * 24.5) / 36.5
= 67 ppm

1854. What is the wavelength of the sound emitted by a guitar string that vibrates 452 times per second in a room where the temperature is 20ºC? (velocity of sound at 20ºC = 1,130 ft/sec)

1) 0.4 ft
2) 0.8 ft
3) 1.2 ft
4) 2.5 ft

Ans.4 If the frequency of a vibrating object is known, the wavelength of the sound it emits bears the following relationship:
Wavelength = Velocity of sound / Frequency
= (1,130 ft/sec) / (452 cycles/sec)
= 2.5 ft

1855. Radiation is the primary way in which energy is transferred:

1) From a beverage to the ice cubes floating in it
2) From a soldering iron to copper wires being soldered
3) From the sun to an earth satellite
4) From a gas stove flame to a pan of water
5)

Ans.3 As the space between the sun and an earth satellite is an almost perfect vacuum and conduction and convection cannot transfer energy in a vacuum, radiation becomes the main source of energy transfer.

1856. A temperature of 120ºC is equivalent to a temperature of:

1) 35° K
2) 153° K
3) 248° K

4) 393° K

Ans.4 Kelvin scale is an absolute thermodynamic scale on which the temperature measured is based on the average kinetic energy per molecule of a perfect gas. The zero on the Kelvin scale is -273.18º Celsius. The size of each Kelvin degree is the same as the Celsius degree. Thus, by adding 273.18 to any Celsius temperature, we obtain the corresponding temperature on the absolute scale or (120 + 273.18)K = 393.18° K.

1857. On the Fahrenheit scale, a temperature of 55ºC is recorded as:

1) 87ºF
2) 119ºF
3) 131ºF
4) 99ºF

Ans.3 Combining the facts that 180 divisions on the Fahrenheit scale are equal to 100 divisions on the Celsius scale (making each Fahrenheit degree 5/9 as large as a Celsius degree) and 0ºC is the equivalent of 32ºF derive the following relationships:
F = 9/5C + 32 and
C = 5/9 (F-32)
Therefore, F = 9/5 (55) + 32 = 131ºF

1858. What would be the effect on a magnet if it were heated with a soldering iron?

1) No effect
2) It would strengthen it
3) It would weaken it
4) It would reverse the polarity

Ans.3 Heating increases the random motion of the particles in the magnet. The arrangement of molecular magnets becomes less orderly, making the magnet weaker.

1859. In a light wave:

1) The vibrations of the electric field are transverse and those of the magnetic field are longitudinal
2) The vibrations of the electric and magnetic fields are transverse
3) All vibrations are longitudinal
4) The vibrations of the magnetic field are transverse and those of the electric field are longitudinal

Ans.2 Light is an electromagnetic wave. In all electromagnetic waves, the electric and magnetic fields vibrate at right angles (transverse) to the direction in which the wave is traveling.

1860. A trait of alpha particles, which are emitted from radioactive nuclei, is that they:

1) Usually consist of protons
2) Usually consist of electrons
3) Are helium nuclei
4) Are hydrogen nuclei

Ans.3 Alpha particles emitted from radioactive sources are helium nuclei. These particles consist of a cluster of two protons and two neutrons (giving a mass number of four, being structurally the same as the nucleus of a helium atom). Alpha particles, on slowing down, combine with electrons from the material through which they are passing and thus become helium atoms.

1861. Neutrons are known to penetrate matter readily mainly because:

1) They are much smaller in size than electrons or protons
2) They have less mass than either electrons or protons
3) They are electrically neutral
4) They possess more energy than electrons or protons

Ans.3 Neutrons are electrically neutral. Compared with electrons and protons their volume is almost the same. The mass of a neutron is slightly more than that of a proton. Characteristically, atoms contain much empty space which neutrons can travel through

without experiencing electric forces from electrons and nuclei because they are electrically neutral.

1862. The greater the frequency of sound waves:

1) The lower the pitch of the sound
2) The louder the sound
3) The softer the sound
4) The higher the pitch of the sound

Ans.4 Frequency is directly related to pitch. The higher pitched a sound is, the higher is its frequency, conversely, low pitched sounds have low frequencies.

1863. An experiment involving sound requires the use of an oscilloscope. On the screen, the loudness of sound should be viewed as a function of:

1) Wavelength
2) Frequency
3) Amplitude
4) Weighted sound level

Ans.3 The intensity of sound is characterized by its loudness, e.g., the louder the sound, the greater the amplitude of its wave motion. On the oscilloscope, the sound wave would appear as an alternating wave motion whose amplitude adjusts to the sound volume.

1864. If a maintenance person moves at a speed of 0.5 times that of sound in a direction away from a noise source producing a sound frequency of f, he would probably hear a sound frequency:

1) Less than f
2) Equal to f
3) 1.5 times greater than f
4) Twice that of f

Ans.1 This is an example of the Doppler effect. Since the maintenance person is moving away from the source of the sound, successive compressions will reach him at the same rate as when he is stationary. Because his rate of speed is less than sound, the compressions don't

catch up with him until the amplitude is too small to have any effect hence, the frequency he hears is less than f.

1865. Sound is a pressure phenomenon. If the sound pressure is doubled, how is the sound level (in decibels) affected?

1) Increases by approximately 6 units
2) Doubles
3) Halved
4) Increases exponentially

Ans.1 Number of dB = 20 log (P_1/P_2)
$P_2 = 2P_1$
dB = 20 log (2/1)
= 6
P_1 and P_2 are the two sound pressure levels.

1866. The average power density one meter from a microwave antenna is 160 mW/cm^2. What is the power density at 2 meters?

1) 160 mW/cm^2
2) 80 mW/cm^2
3) 60 mW/cm^2
4) 40 mW/cm^2

Ans.4 Use the inverse square law. PWD = 160 $mW/cm^2(1\ m / 2\ m)^2$) = 40 mW/cm^2

1867. Transmission loss (TL) is defined by any of the following EXCEPT:

1) Energy lost as sound travels through a material
2) Is a ratio of two numbers
3) Affected by openings in an acoustical barrier
4) Reflective property of a material

Ans.4 Transmission loss is defined as 10 times the log of the ratio of the acoustic energy transmitted through a barrier to the incident acoustic energy. It is a physical property of the material used for a given construction.

1868. What is the smallest particle size that can be seen with the unaided eye?

1) 1 - 10 αm
2) 10 - 50 αm
3) 50 - 100 αm
4) 100 - 200 αm

Ans.3 50 - 100 ⌠m is the smallest particle size that can be seen unaided by the human eye.

1869. The most common source of radon gas is:

1) Soil
2) Building materials
3) Well water
4) Carpeting

Ans.1 Soil and bedrock that is rich in uranium or radium will off-gas radon, which can enter a home basement through cracks or other openings. Building materials only rarely contribute to radon gas in a home, unless made with contaminated materials. Well water can contribute to radon gas exposure, but only at very high water concentrations. Carpeting is not a source of radon.

1870. The half-life of radon is:

1) 1 year
2) 38 days
3) 3.8 days
4) 1 day

Ans.3 The half-life of radon gas is 3.8 days.

1871. Radon emits what type(s) of radiation when it decays?

1) Alpha
2) Beta
3) Gamma
4) Alpha and gamma

Ans.4 Radon 222 produces alpha particles and gamma radiation when it decays to polonium 218.

1872. Radon is a hazard for all of the following reasons EXCEPT:

1) Potential for lung cancer
2) Birth defects from exposure of pregnant women in first trimester
3) Exposure to decay products from inhalation of dust
4) Decay products trapped in lung decay and damage lung tissue

Ans.2 Radon is not known to cause birth defects. It is known to cause lung cancer from inhalation of dust particles contaminated with radon decay products, which further decay and can cause lung damage.

1873. The disintegration rate of 1 gram of radium is 3.7×10^{10} d/s. This rate of decay has units of:

1) Half-life
2) Fuel-life
3) Average rate
4) Curie

Ans.4 A curie is a unit of rate of decay.

1874. Biohazards can be transmitted to a person by:

1) Inhalation or accidental injection
2) Inhalation, ingestion, or absorption
3) Physical contact
4) All of the above

Ans.4 Biohazards can be transmitted to a person through inhalation of airborne agents, injection (usually accidental), ingestion through contaminated food or water, and absorption by physical contact with contaminated surfaces or clothing.

1875. In toxicology, we show concern for the inhalation of dust particles measured in microns. What is the size of a micron?

1) Approximately l/25,000 of an inch
2) 0.0001 millimeters
3) 10^{-4} centimeters
4) Both 1 and 3

Ans.4 A micron is defined as a unit of length, the thousandth part of a millimeter (0.001 mm) or the millionth part of a meter (10^{-6} m or 10^{-4} cm) which is approximately 1/25,000 of an inch.

1876. A total flow of 800 cfm was measured in a large duct 5 ft * 5 ft. What is the average velocity of the air at the entrance of the duct?

1) 2000 ft/min
2) 160 cfm
3) 32 ft/min
4) 32 cfm

Ans.3 Q = AV
A = 5 ft * 5 ft = 25 ft^2
Q = 800 cfm
800 cfm = (25 sq ft)V
V = (800 ft^3/min) / 25 ft^2
V = 32 ft/min

1877. Given a velocity pressure reading of 0.60 in. of water in a circular duct, what is the average velocity in ft/min?

1) 15 ft/min
2) 29 ft/min
3) 1,023 ft/min
4) 3,102 ft/min

Ans.4

$$V = 4005\sqrt{VP}$$

Where:
VP = velocity pressure

$$V = 4005\sqrt{0.60}$$

= 3,102 ft/min

1878. If 3,500 cfm of air is drawn through an 8 in. diameter duct, what is the velocity in ft/min?

1) 800 ft/min
2) 1,000 ft/min
3) 1,023 ft/min
4) 10,000 ft/min

Ans.4 Q = VA
A = □(0.33 ft)^2
= 0.349 ft^2
V = Q/A = (3,500 ft^3/min) / 0.349 ft^2
= 10,027 ft/min

1879. In a mine opening you found a total flow of 2,000 cfm. The opening was 10 ft * 10 ft. What is the velocity of the air in the entrance?

1) 2 ft/min
2) 3 ft/min
3) 20 ft/min
4) 3,000 ft/min

Ans.3 Q = VA
V = Q/A
V = 2,000 cfm / 100 ft^2
= 20 ft/min

1880. In industrial ventilation system design, which of the following assumptions is FALSE?

1) Air is considered a compressible fluid
2) Air flow requirements should be minimized
3) Energy should be conserved
4) The worker's breathing zone must be protected
5)

Ans.1 Air is considered an incompressible fluid. This assumption allows various laws to be applied which otherwise could not be used to estimate air flows, velocities, etc.

1881. The function of a slot in a slot hood is to:

1) Decrease capture velocity
2) Reduce material required to build system
3) Obtain proper air distribution
4) Increase capture velocity

Ans.3 Capture velocity is essentially independent of its slot air velocity. The function of the slot is to provide proper air distribution by making the pressure drop of the slot large relative to the plenum.

1882. The rule of thumb for determining the average duct velocity from the center point velocity is to multiply the center point velocity by:

1) 0.75
2) 0.90
3) 1.10
4) No generalization can be made

Ans.2 When accuracy is not the prime concern, a velocity measurement can be made at least 10 diameters downstream and the center line velocity multiplied by 0.90. If VP is measured, then multiply VP * 0.81.

1883. Total pressure is:

1) The difference between SP and VP
2) Measured parallel to the axis of flow
3) Normally positive on the suction side
4) Measured perpendicular to the axis of flow

Ans.2 It is measured parallel to the direction of flow since it contains a velocity pressure component. Total pressure is equal to the sum of the static pressure (SP) and the velocity pressure (VP), not the difference.

1884. Velocity pressure of a system is always:

1) Negative
2) Positive
3) Equal to total pressure
4) Equal to static pressure

Ans.2 The total pressure = static pressure + velocity pressure. Since VP is always exerted in the direction of flow, VP is always positive.

1885. What is the duct velocity in a 6 in. duct with an airflow of 1,000 cfm?

1) 5,000 ft/min
2) 5,092 ft/min
3) 6,750 ft/min
4) 7,225 ft/min

Ans.2 The area of a 6 in duct is 0.1964 ft^2
(remember to change inches to feet when calculating the area of the duct).
Start with Q = AV and solve for V.
V = Q/A = (1,000 ft^3/min) / 0.1964 ft^2
= 5,092 ft/min

1886. What is the recommended duct velocity for an average industrial dust such as wool, wood, sand abrasive blasting dust, or shoe dust?

1) 2,000 ft/min
2) 3,000 ft/min
3) 4,000 ft/min
4) 5,000 ft/min

Ans.3 4,000 ft/min is value is recommended by the ACGIH in the Industrial Ventilation Handbook.

1887. Which of the following is FALSE?

1) Q varies indirectly as fan speed
2) SP varies as the square of fan speed
3) HP varies as the cube of the fan speed
4) TP varies as the square of fan speed

Ans.1 Q varies directly as fan speed and not indirectly. The other listed relations between some operating parameter and the fan are true.

1888. Which of the following is NOT a characteristic of a centrifugal fan?

1) High volume - low pressure drop
2) Low space requirements
3) Often used with particle-laden air
4) Low to medium noise

Ans.1 High volume - low pressure drop are the most notable characteristics of axial flow fans, NOT radial or centrifugal flow fans.

1889. Which of the following statements concerning various types of air pressure is NOT true?

1) Velocity pressure may be a positive or negative number
2) Static pressure may be a positive or negative number
3) SP is exerted on the sides of the duct
4) TP = SP + VP

Ans.1 Velocity pressure is a measure of kinetic energy and therefore cannot be negative. Static pressure can be negative if the duct is an exhaust duct.

1890. Which of the following statements is NOT true about the static pressure in a given ventilation system?

1) Is measured parallel to the direction of flow
2) Tends to collapse the duct in an exhaust system
3) Acts in all directions
4) Is available to do work

Ans.1 The static pressure is always measured perpendicular to the direction of flow.

1891. Why would 4,000 ft/min be maintained in an exhaust duct?

1) Prevents settling

2) Reduces abrasion
3) Minimizes noise
4) Minimizes energy consumption

Ans.1 A velocity this high would normally be used only if dust transport were involved.

1892. The basic ionizing radiation control measures include any of the following EXCEPT:

1) Minimizing time of exposure
2) Source substitution
3) Using shielding
4) Increasing distance

Ans.2 The basic philosophy in reducing ionizing radiation exposures is time, distance, and shielding. Keep the time of exposure as low as possible, increase the distance as much as possible, and shield the source with an effective shielding material.

1893. Which radioactive emission is the easiest to shield against?

1) Alpha
2) Beta
3) Neutron
4) Gamma

Ans.1 Alpha particles are relatively easy to shield. Paper is an effective alpha shield. The other answers (beta, neutron, and gamma) are more difficult to shield.

1894. Why are the low-atomic-number materials such as aluminum and organo-plastics used to shield beta particles?

1) Cheaper to use
2) Reduces Bremmsstrahlung radiation
3) Higher atomic number materials are not effective
4) Lighter than lead

Ans.2 Beta particles can generate Bremmsstrahlung (braking radiation) X-rays when they are slowed too quickly by hitting materials of high atomic numbers. Therefore, aluminum and plastics with atomic numbers less than 14 are utilized for shielding beta particles.

1895. What type of shielding is used for X-ray or gamma radiation?

1) Paper
2) Aluminum
3) Plastic
4) Lead

Ans.4 This type of higher energy radiation can most effectively be shielded by lead or concrete. The other materials, while effective for other types of radiation, would not be effective for X-ray or gamma radiation shielding.

1896. In which of the following wavelength groupings is the lowest amount of radiant exposure allowed?

1) 200 - 220 nm
2) 220 - 240 nm
3) 240 - 260 nm
4) 260 - 280 nm

Ans.4 The lowest allowable radiant energy exposures occur in the region 260 - 280 nm (nanometers). This is the UV region where exposures can cause the most damage. Note that allowable exposure INCREASES at around 270 nm. The 260 - 280 region is a minimum on the curve. Review this graph in the ACGIG TLVs and BEIs publication.

1897. What type of respiratory protection is minimally required for Level C protection?

1) NIOSH-approved full or half-face air-purifying respirator
2) NIOSH-approved full-face piece respirator only
3) NIOSH-approved SCBA
4) NIOSH-approved Positive pressure SCBA only

Ans.1 Level C protection includes full-face or half-face air purifying respirator.
29 CFR 1910.120 Appendix B

1898. The following are types of glove materials EXCEPT:

1) Nitrile
2) Neoprene
3) PVC
4) Polynitrile

Ans.4 Polynitrile is not a glove material. Nitrile, PVC, and neoprene are common glove materials.

1899. Before donning full-body chemical protective clothing, one should inspect all the following EXCEPT:

1) Seams
2) Snaps/buttons
3) Pressure relief valves
4) Other openings

Ans.2 Full-body chemical protective clothing does not contain buttons or snaps, otherwise there would be openings for chemicals to enter.

1900. You would use a cartridge respirator during chemical spill cleanup operations when the following conditions apply EXCEPT:

1) The oxygen content is above 19.5%
2) The concentration of airborne material is less than the protection factor for the respirator
3) Hazards and concentrations of materials are known
4) The odor threshold of the material is greater than the TLV

Ans.4 A cartridge respirator cannot be used when the material has poor warning properties. If the odor threshold is greater than the TLV, the warning properties are poor.

1901. If chemical warnings are not accessible when responding to a spill:

1) Odor can be used to determine the level of protection
2) No one should be sent into the spill area
3) Level A protection should be donned
4) The maximum level of respiratory protection and gloves should be donned

Ans.3 Level A protection will offer complete isolation from the hazard to allow safe emergency response.

1902. The most sensitive and accurate method of physiologically measuring work is:

1) Heart rate
2) Oxygen consumption
3) Respiratory volume
4) Electromyography

Ans.2 The body's oxygen consumption is the most sensitive and accurate method of measuring work.

1903. The sound pressure unit "dB" has become popular because:

1) It is specific to noise and its action
2) It covers a wide range of values
3) It is the only term available
4) It is easy to combine various sources

Ans.2 The dB is a logarithmic measure and therefore can cover a wide range of values. It would be cumbersome to use numbers which vary by a factor of more than one million.

1904. What control measures can be taken for adequate protection against external radiation hazards?

1) Use lead shielding and protective equipment
2) Follow the principals of the inverse square law
3) Limit exposure time, work at a safe distance, and use barriers
4) Use written procedures, employee training, personal protective equipment, and signage

Ans.3 Gamma rays, X-rays and neutrons are the most penetrating radiations and usually constitute the major external hazards. There are three principal factors that provide the basis for radiation protection: time, distance, and shielding.

TIME - To control personnel exposure, the dose rate from a radiation source may be measured and appropriate limits placed on the working time for control of the dose to the individual.

DISTANCE - The intensity of a radiation field decreases as you move further away from the radiation source. The relation between radiation exposure rate and distance follows the inverse square law.

SHIELDING - Since ionizing radiation will interact with most materials, in principle any can be used for shielding. Some materials, however, like lead, make more efficient shields than do others.

1905. The basic color used for designating radiation hazards is:

1) Yellow
2) Orange
3) Blue
4) Purple

Ans.4 Purple is the basic color for designating radiation hazards. Examples include: Radiation warning signs, containers of radioactive material, and signal lights to indicate radiation machines are in operation. The safety color code for marking physical hazards and the identification of certain equipment is detailed in ANSI Z53.l and 29 CFR 1910.144.